

LSE Module 2 – Assignment Report

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0. Business Rationale:

The NHS has provided a business case to analyze current service demands and utilization rates in response to growing populations and demands. The key is balancing adequate capacity with efficient utilization of existing resources. The accompanying data sheds light on appointment distribution across different regions, attendance rates, workload levels, and appointment wait times. Additionally, social media insights from Twitter help capture trending discussions, offering valuable context & data-driven insights that could inform the NHS's future infrastructure and budget decisions

1. Analytical Approach

Our approach to analyzing the NHS data involved several stages, from initial data exploration and cleaning to detailed analysis and sentiment assessment of tweets.

1.1a Data Exploration and Loading

We started by loading four key datasets(named as in brackets):

appointments_regional.csv (ar) ,
actual_duration.csv (ad),
national_categories.xlsx (nc), and tweets.csv (tweets).

Each dataset contained different but complementary information, such as appointment distributions, service settings, and social media insights. After loading the datasets into separate DataFrames (ar, ad, nc, and tweets), we inspected their structure (Fig1.0), checked for missing values, and examined the columns (Fig 1.1 -1.4) to understand what each dataset contained (Ref Appendix 1.1 for detailed code).

```
********************************** The Shapes of the Data Frames **********************************
Appointments Regional -----  (596821, 7)
Actual Duration -----  (137793, 8)
National Category -----  (817394, 8)
Tweets -----  (1174, 10)
```

Fig 1.0

```
****************************** Columns, Non Null Counts of Dataframes ***************************

---------------------- Appointments Regional ----------------------

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 596821 entries, 0 to 596820
Data columns (total 7 columns):
 #   Column                            Non-Null Count   Dtype
---  ------                            --------------   -----
 0   icb_ons_code                      596821 non-null  object
 1   appointment_month                 596821 non-null  object
 2   appointment_status                596821 non-null  object
 3   hcp_type                          596821 non-null  object
 4   appointment_mode                  596821 non-null  object
 5   time_between_book_and_appointment  596821 non-null  object
 6   count_of_appointments             596821 non-null  int64
dtypes: int64(1), object(6)
memory usage: 31.9+ MB
None

----------------------- Actual Duration -------------------------

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 137793 entries, 0 to 137792
Data columns (total 8 columns):
 #   Column                     Non-Null Count   Dtype
---  ------                     --------------   -----
 0   sub_icb_location_code      137793 non-null  object
 1   sub_icb_location_ons_code  137793 non-null  object
 2   sub_icb_location_name      137793 non-null  object
 3   icb_ons_code               137793 non-null  object
 4   region_ons_code            137793 non-null  object
 5   appointment_date           137793 non-null  object
 6   actual_duration            137793 non-null  object
 7   count_of_appointments      137793 non-null  int64
dtypes: int64(1), object(7)
memory usage: 8.4+ MB
None
```

Fig 1.1

```
----------------------- National Category -------------------------

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 817394 entries, 0 to 817393
Data columns (total 8 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   appointment_date       817394 non-null  datetime64[ns]
 1   icb_ons_code           817394 non-null  object
 2   sub_icb_location_name  817394 non-null  object
 3   service_setting        817394 non-null  object
 4   context_type           817394 non-null  object
 5   national_category      817394 non-null  object
 6   count_of_appointments  817394 non-null  int64
 7   appointment_month      817394 non-null  object
dtypes: datetime64[ns](1), int64(1), object(6)
memory usage: 49.9+ MB
None

---------------------- Tweets -------------------------

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1174 entries, 0 to 1173
Data columns (total 10 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   tweet_id                 1174 non-null   int64
 1   tweet_full_text          1174 non-null   object
 2   tweet_entities           1174 non-null   object
 3   tweet_entities_hashtags  1007 non-null   object
 4   tweet_metadata           1174 non-null   object
 5   tweet_retweet_count      1174 non-null   int64
 6   tweet_favorite_count     1174 non-null   int64
 7   tweet_favorited          1174 non-null   bool
 8   tweet_retweeted          1174 non-null   bool
 9   tweet_lang               1174 non-null   object
dtypes: bool(2), int64(3), object(5)
memory usage: 75.8+ KB
```

Fig 1.2

To ensure that the data was ready for analysis, we obtained non-null counts and general information about each DataFrame (Fig 1.1-1.2). This helped identify any missing values or inconsistencies, especially in the categorical columns, which informed subsequent data-cleaning steps.

We found statistics for each data frame depending on whether

1. it contained appointment-related or geographical information
2. it had numerical or categorical columns.

1.1.b Appointment related information:

Numerical -

1. Count of Appointments – from Appointments Regional, Actual Duration, National Category (Fig 1.3)
2. Tweets Retweet Count & Tweet Favorite Count - from Tweets (Fig 1.4)

Categorical -

1. Appointment Status, Health Care Professional (hcp_type), Appointment mode & Time between Booking and Appointment – from Appointments Regional (Fig 1.5)

2. Actual Duration – from Actual Duration (Fig 1.6)

3. Service Setting, Context Type & National Category – from National Category (Fig 1.7)

```
****************************** Statistics of Count of Appointments **************************

---------------------- Appointments Regional ----------------------

       count_of_appointments
count          596821.000000
mean             1244.601857
std              5856.887042
min                 1.000000
25%                 7.000000
50%                47.000000
75%               308.000000
max            211265.000000

----------------------- Actual Duration --------------------------

       count_of_appointments
count          137793.000000
mean             1219.080011
std              1546.902956
min                 1.000000
25%               194.000000
50%               696.000000
75%              1621.000000
max             15400.000000

---------------------- National Category -------------------------

       count_of_appointments
count          817394.000000
mean              362.183684
std              1084.576600
min                 1.000000
25%                 7.000000
50%                25.000000
75%               128.000000
max             16590.000000
```

Fig 1.3

```
 ----------------------- Tweets -------------------------------------

        tweet_retweet_count  tweet_favorite_count
 count          1174.000000            1174.00000
 mean              8.629472               0.37138
 std              29.784675               2.04470
 min               0.000000               0.00000
 25%               0.000000               0.00000
 50%               1.000000               0.00000
 75%               3.000000               0.00000
 max             303.000000              42.00000
```

Fig 1.4

```
******************************** Statistics of Appointments Related Info ***************************

---------------------- Appointments Regional --------------------

|| appointment_status || --> Categories & Counts

Attended    232137
Unknown     201324
DNA         163360
Name: appointment_status, dtype: int64

|| hcp_type || --> Categories & Counts

Other Practice staff    241557
GP                      226036
Unknown                 129228
Name: hcp_type, dtype: int64

|| appointment_mode || --> Categories & Counts

Face-to-Face    180410
Telephone       166483
Home Visit      111194
Unknown          79147
Video/Online     59587
Name: appointment_mode, dtype: int64
```

Fig 1.5

```
|| time_between_book_and_appointment || --> Categories & Counts

Same Day                  95502
2 to 7 Days               92409
1 Day                     88957
8  to 14 Days             82698
15  to 21 Days            73666
22  to 28 Days            68755
More than 28 Days         65147
Unknown / Data Quality    29687
Name: time_between_book_and_appointment, dtype: int64


------------------------ Actual Duration --------------------------

|| actual_duration || --> Categories & Counts

Unknown / Data Quality    20161
1-5 Minutes               19909
6-10 Minutes              19902
11-15 Minutes             19738
16-20 Minutes             19534
21-30 Minutes             19452
31-60 Minutes             19097
Name: actual_duration, dtype: int64
```

Fig 1.6

```
---------------------- National Category --------------------------

|| service_setting || --> Categories & Counts

General Practice             359274
Primary Care Network         183790
Other                        138789
Extended Access Provision    108122
Unmapped                      27419
Name: service_setting, dtype: int64

|| context_type || --> Categories & Counts

Care Related Encounter    700481
Inconsistent Mapping       89494
Unmapped                   27419
Name: context_type, dtype: int64

|| national_category || --> Categories & Counts

Inconsistent Mapping                                                89494
General Consultation Routine                                        89329
General Consultation Acute                                          84874
Planned Clinics                                                     76429
Clinical Triage                                                     74539
Planned Clinical Procedure                                          59631
Structured Medication Review                                        44467
Service provided by organisation external to the practice           43095
Home Visit                                                          41850
Unplanned Clinical Activity                                         40415
Patient contact during Care Home Round                              28795
Unmapped                                                            27419
Care Home Visit                                                     26644
Social Prescribing Service                                          26492
Care Home Needs Assessment & Personalised Care and Support Planning 23505
Non-contractual chargeable work                                     20896
Walk-in                                                             14179
Group Consultation and Group Education                               5341
```

Fig 1.7

1.1.c Geographical related information:

Categorical-

1. ICB code – from Appointments Regional (Fig 1.8)
2. ICB, Sub ICB, location names, region from Actual Duration (Fig 1.9, 1.10,1.11)
3. ICB, Sub ICB from National Category (Fig. 1.11, 1.12)

```
****************************** Statistics of Appointments Related Info ***************************

--------------------- Appointments Regional --------------------

|| icb_ons_code || --> Categories & Counts

E54000057    54527
E54000008    48253
E54000050    43774
E54000048    36360
E54000051    31069
E54000010    27401
E54000054    26960
E54000061    23466
E54000026    22342
E54000044    19806
E54000053    18510
E54000025    16809
E54000023    16326
E54000015    15587
E54000042    12057
E54000060     9608
E54000055     8206
E54000029     7700
E54000032     7615
E54000062     7561
E54000030     7462
E54000037     7453
E54000031     7324
E54000027     7299
E54000056     7178
E54000039     6993
E54000052     6921
E54000058     6875
E54000038     6813
E54000036     6714
E54000043     6707
E54000022     6582
```

Fig 1.8

```
------------------------ Actual Duration --------------------------

|| sub_icb_location_code || --> Categories & Counts

26A      1484
91Q      1484
W2U3Z    1484
M1J4Y    1484
14L      1484
         ...
05D      1051
03L      1028
00P      1025
02P      1022
01F      1013
Name: sub_icb_location_code, Length: 106, dtype: int64

|| sub_icb_location_ons_code || --> Categories & Counts

E38000239    1484
E38000237    1484
E38000256    1484
E38000249    1484
E38000217    1484
             ...
E38000053    1051
E38000141    1028
E38000176    1025
E38000006    1022
E38000068    1013
Name: sub_icb_location_ons_code, Length: 106, dtype: int64
```

Fig 1.9

```
|| sub_icb_location_name || --> Categories & Counts

NHS Norfolk and Waveney ICB - 26A                        1484
NHS Kent and Medway ICB - 91Q                            1484
NHS North West London ICB - W2U3Z                        1484
NHS Bedfordshire Luton and Milton Keynes ICB - M1J4Y     1484
NHS Greater Manchester ICB - 14L                         1484
                                                         ...
NHS Staffordshire and Stoke-on-Trent ICB - 05D           1051
NHS South Yorkshire ICB - 03L                            1028
NHS North East and North Cumbria ICB - 00P               1025
NHS South Yorkshire ICB - 02P                            1022
NHS Cheshire and Merseyside ICB - 01F                    1013
Name: sub_icb_location_name, Length: 106, dtype: int64

|| icb_ons_code || --> Categories & Counts

E54000057    12668
E54000008    10636
E54000050     9938
E54000048     9808
E54000051     7973
E54000010     7208
E54000054     6681
E54000026     6653
E54000061     4471
E54000053     4339
E54000015     4103
E54000044     3930
E54000025     3903
E54000023     3893
E54000060     2561
E54000042     2513
E54000032     1484
E54000024     1484
```

Fig 1.10

```
|| region_ons_code || --> Categories & Counts

E40000010    33112
E40000012    29063
E40000011    25015
E40000007    18888
E40000005    15130
E40000006     9465
E40000003     7120
Name: region_ons_code, dtype: int64


---------------------- National Category -------------------------

|| icb_ons_code || --> Categories & Counts

E54000057    59180
E54000008    53744
E54000050    53707
E54000048    43381
E54000054    42888
E54000051    41402
E54000010    34808
E54000026    33309
E54000061    27812
E54000044    27539
E54000015    27409
E54000053    25835
E54000025    25242
E54000023    20632
E54000042    17419
E54000060    14473
E54000027    13007
E54000032    12637
E54000037    12526
E54000029    11837
```

Fig 1.11

```
|| sub_icb_location_name || --> Categories & Counts

NHS North West London ICB - W2U3Z              13007
NHS Kent and Medway ICB - 91Q                  12637
NHS Devon ICB - 15N                            12526
NHS Hampshire and Isle Of Wight ICB - D9Y0V    12171
NHS North East London ICB - A3A8R              11837
                                               ...
NHS North East and North Cumbria ICB - 00N      4210
NHS Lancashire and South Cumbria ICB - 02G      4169
NHS Cheshire and Merseyside ICB - 01V           3496
NHS Cheshire and Merseyside ICB - 01T           3242
NHS Greater Manchester ICB - 00V                2170
Name: sub_icb_location_name, Length: 106, dtype: int64
```

Fig 1.12

2. Data Cleaning and Processing

2.1.a Processing

To ensure the data was ready for detailed analysis, several cleaning and processing steps were performed (Table 2.1-2.2, Fig 2.1 -2.5 & Appendix 2.1):

Tweets Dataset

Step	Description	Figure Reference
Filtering	We filtered for tweets containing relevant NHS-related hashtags by excluding rows with empty values in tweet_entities_hashtags, resulting in a subset with **1,007 records**. This allowed us to focus on tweets with NHS-specific content.	Fig 1.2
Hashtag Processing and Date Assignment	Hashtags within each tweet were split and exploded to create individual rows for each hashtag. We standardized hashtags to lowercase and trimmed any extra spaces to ensure consistency in analysis.	Fig 2.1
Date Extraction from Twitter IDs	Using Twitter IDs, we derived the posting dates, enabling chronological analysis of tweets. This step facilitated the exploration of time-based trends in tweet activity.	Fig 2.2
Sentiment Analysis	Utilizing the **NLTK** and **VADER Sentiment** libraries, we analyzed the sentiment of tweets, categorizing them as **positive**, **negative**, or **neutral**. Prior to sentiment scoring, text preprocessing was performed to remove URLs, special characters, and excess whitespace, standardizing the text for effective analysis.	Fig 2.3
Keyword Matching for Relevant Tweets	We identified relevant tweets containing specific keywords and phrases related to NHS and healthcare topics. A list of keywords, including concatenated versions (e.g., "healthcareprofessional" vs. "healthcare professional"), was created. We applied regex to detect these keywords in tweet texts and filtered tweets to retain only those containing at least one relevant keyword, enabling a focused analysis of healthcare-related themes.	Fig 2.4

Table 2.1

	tweet_id	tweet_full_text	tweet_entities	tweet_entities_hashtags	hashtags_list	hashtags_list_lower
0	1567629223795527681	As Arkansas' first Comprehensive Stroke Certif...	{'hashtags': [{'text': 'Healthcare', 'indices'...	#Healthcare	#Healthcare	#healthcare
2	1567582787070304256	RT @OntarioGreens: $10 billion can go a long w...	{'hashtags': [{'text': 'Healthcare', 'indices'...	#Healthcare	#Healthcare	#healthcare
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	{'hashtags': [{'text': 'blogs', 'indices': [18...	#blogs, #digitaltransformation, #cybersecurity...	#blogs	#blogs
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	{'hashtags': [{'text': 'blogs', 'indices': [18...	#blogs, #digitaltransformation, #cybersecurity...	#digitaltransformation	#digitaltransformation
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	{'hashtags': [{'text': 'blogs', 'indices': [18...	#blogs, #digitaltransformation, #cybersecurity...	#cybersecurity	#cybersecurity

Fig 2.1

	tweet_id	tweet_full_text	Date
0	1567629223795527681	As Arkansas' first Comprehensive Stroke Certif...	2022-09-07
2	1567582787070304256	RT @OntarioGreens: $10 billion can go a long w...	2022-09-07
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	2022-09-07
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	2022-09-07
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	2022-09-07

Fig 2.2

	tweet_id	tweet_full_text	tweet_entities_hashtags	hashtags_list	sentiment_scores	compound_score	sentiment
0	1567629223795527681	As Arkansas' first Comprehensive Stroke Certif...	#Healthcare	#Healthcare	{'neg': 0.0, 'neu': 0.768, 'pos': 0.232, 'comp...	0.8384	positive
2	1567582787070304256	RT @OntarioGreens: $10 billion can go a long w...	#Healthcare	#Healthcare	{'neg': 0.134, 'neu': 0.866, 'pos': 0.0, 'comp...	-0.4767	negative
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	#blogs, #digitaltransformation, #cybersecurity...	#blogs	{'neg': 0.0, 'neu': 0.814, 'pos': 0.186, 'comp...	0.6249	positive
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	#blogs, #digitaltransformation, #cybersecurity...	#digitaltransformation	{'neg': 0.0, 'neu': 0.814, 'pos': 0.186, 'comp...	0.6249	positive
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	#blogs, #digitaltransformation, #cybersecurity...	#cybersecurity	{'neg': 0.0, 'neu': 0.814, 'pos': 0.186, 'comp...	0.6249	positive

Fig 2.3

	tweet_id	tweet_full_text	tweet_full_text_lower	found_keywords
0	1567629223795527681	As Arkansas' first Comprehensive Stroke Certif...	as arkansas' first comprehensive stroke certif...	[health, healthcare]
2	1567582787070304256	RT @OntarioGreens: $10 billion can go a long w...	rt @ontariogreens: $10 billion can go a long w...	[healthcare]
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	icymi: our recent blogs on cybersecurity in ac...	[healthcare]
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	icymi: our recent blogs on cybersecurity in ac...	[healthcare]
4	1567582720460570625	ICYMI: Our recent blogs on Cybersecurity in Ac...	icymi: our recent blogs on cybersecurity in ac...	[healthcare]
...	...	...	...	...

Fig 2.4

Other Datasets

Dataset	Description	Figure Reference
Appointments Regional (ar)	We formatted the appointment_month to pandas datetime format (appointment_month_fr). From the date fields, we created **quarter** and **month** columns. These enhancements enabled time-series analysis, making it easier to aggregate and compare appointment counts over specific time periods and by Integrated Care Board (ICB) codes.	Fig 2.5
Actual Duration (ad)	Similar to the appointments dataset, we formatted date fields and extracted time components to facilitate chronological analysis.	Fig 2.6

Table 2.2

	appointment_month_fr	appointment_quarter
0	2020-01-01	2020Q1
1	2020-01-01	2020Q1
2	2020-01-01	2020Q1
3	2020-01-01	2020Q1
4	2020-01-01	2020Q1

Fig 2.5

	appointment_date_fr	appointment_month	appointment_quarter
0	2021-12-01	2021-12	2021Q4
1	2021-12-01	2021-12	2021Q4
2	2021-12-01	2021-12	2021Q4
3	2021-12-01	2021-12	2021Q4
4	2021-12-01	2021-12	2021Q4

Fig 2.6

3.Data Wrangling & Transformation for Visualization

To facilitate visual analysis, we performed a series of wrangling and transformation steps on the *Appointments Regional*, *Actual Duration*, *National Category* & *Tweet* (Fig 3.1-3.10). The objective is to structure the data for easy comparison of appointment status counts across two categorical variables (Figs 1.5-1.12).

3.1.a Internal data - Appointments

For *Appointments Regional*, *Actual Duration*, *National Category* Table 3.1 outlines the steps we follow for data wrangling and transformation.

Step	Description	Figure Reference
1. Filter Data	Remove rows with irrelevant labels (e.g., "Unknown," "Missing") to focus on meaningful data.	Fig 3.1 - Fig 3.3
2. Group by Variables	Aggregate data by grouping and calculating total appointment counts for combinations of two categorical variables (e.g., appointment_mode and appointment_status).	Fig 3.1 - Fig 3.3
3. Calculate Totals per Category	Determine overall counts for each level of the baseline variable (e.g., appointment_mode) as a basis for percentage calculation.	Fig 3.1 - Fig 3.3
4. Calculate Percentages	Calculate the percentage of each category level within each baseline variable level, normalizing data for cross-category comparison.	Fig 3.1 - Fig 3.3
5. Pivot for Visualization	Reshape the data with baseline categories as rows and secondary categories as columns, preparing it for stacked visualization.	Fig 3.2 - Fig 3.4

Table 3.1

3.1.b Internal Data – Geographies

For appointment- & geographical related insights, count of appointments were found for combination of categorical variables & standalone categorical variable as seen in below Table 3.2.

Type of Insight	Dataset	Category	Standalone Variable	Combinations with Other Variables
Appointment-Related	Appointments Regional	Appointment Status	appointment_status	- appointment_status and appointment_mode - appointment_status and hcp_type - appointment_status and time_between_book_and_appointment - appointment_status and appointment_month
		Healthcare Professional	hcp_type	- hcp_type and appointment_mode - hcp_type and time_between_book_and_appointment - hcp_type and appointment_month
		Appointment Mode	appointment_mode	- appointment_mode and time_between_book_and_appointment - appointment_mode and appointment_month
		Time Between Booking & Appointment	time_between_book_and_appointment	- time_between_book_and_appointment and appointment_month
	Actual Duration	Actual Duration	actual_duration	- actual_duration and appointment_month
	National Category	National Category	national_category	- national_category and service_setting - national_category and appointment_month
		Service Setting	service_setting	- service_setting and appointment_month
Geographical-Related	Appointments Regional	ICB ONS Code	icb_ons_code	- icb_ons_code and appointment_mode - icb_ons_code and appointment_status - icb_ons_code and hcp_type - icb_ons_code and time_between_book_and_appointment
	Actual Duration			- icb_ons_code and actual_duration
	National Category			- icb_ons_code and national_category - icb_ons_code and service_setting

Table 3.2 Appointment & Geographical Insights

3.1.c External Data - Tweets

For *Tweets,* Table 3.3 outlines the steps we follow for data wrangling and transformation.(ref Appendix 3.1& 3.2)

Step	Description	Figure Reference
1. Group Hashtags	Grouped the tweets_filtered DataFrame by hashtags_list_lower and summed the tweet_retweet_count for each hashtag.	Fig 3.5
2. Remove Zero Retweets	Removed hashtags with zero retweets and sorted the remaining hashtags in descending order based on their total retweet counts.	Fig 3.5
3. Categorize Hashtags	Categorized hashtags into levels (e.g., High_retweet_hash, First_retweet_hash) based on their retweet performance for targeted sentiment analysis.	Fig 3.6
4. Identify Engaged Hashtags	Identified the most engaged hashtags and their sentiments, facilitating focused and meaningful data visualization.	Fig 3.5

Table 3.3 Data Wrangling Steps forTwitter Data

	appointment_mode	appointment_status	count_of_appointments	total_per_mode	percentage
0	Face-to-Face	Attended	392705999	416959037	94.183352
1	Face-to-Face	DNA	24253038	416959037	5.816648
2	Home Visit	Attended	3612068	3799639	95.063452
3	Home Visit	DNA	187571	3799639	4.936548
4	Telephone	Attended	254799226	260061477	97.976536
5	Telephone	DNA	5262251	260061477	2.023464
6	Video/Online	Attended	3303233	3462179	95.409076
7	Video/Online	DNA	158946	3462179	4.590924

Fig 3.1 Appointment Mode/Status Group by

	count_of_appointments		percentage	
appointment_status	Attended	DNA	Attended	DNA
appointment_mode				
Face-to-Face	392705999.0	24253038.0	94.183352	5.816648
Home Visit	3612068.0	187571.0	95.063452	4.936548
Telephone	254799226.0	5262251.0	97.976536	2.023464
Video/Online	3303233.0	158946.0	95.409076	4.590924

Fig 3.2 Appointment Mode/Status Regional Pivot

	hcp_type	appointment_status	count_of_appointments	total_per_hcp	percentage
0	GP	Attended	357847005	367275940	97.432738
1	GP	DNA	9428935	367275940	2.567262
2	Other Practice staff	Attended	299129577	319958912	93.489997
3	Other Practice staff	DNA	20829335	319958912	6.510003

Fig 3.3 Healthcare Professional Type/Appointment Mode Group by

	count_of_appointments		percentage	
appointment_status	Attended	DNA	Attended	DNA
hcp_type				
GP	357847005.0	9428935.0	97.432738	2.567262
Other Practice staff	299129577.0	20829335.0	93.489997	6.510003

Fig 3.4 Healthcare Professional Type/Appointment Mode Pivot

	hashtags_list_lower	tweet_retweet_count
547	#healthcare	3190
39	#ai	875
542	#health	790
374	#digitalhealth	735
577	#healthtech	611
...	...	...
849	#mobilehealth	1
859	#msftadvocate	1
862	#n95	1
885	#news_音楽	1
1443	#zoom	1

Fig 3.5 Hashtag ReTweet in Descending order

```
High_retweet_hash = [
    '#healthcare'
]

First_retweet_hash = [
    '#health', '#healthtech', '#meded', '#medical', '#medicine', '#medtwitter', '#medicare',
    '#telehealth', '#clinicalpearl', '#clinicaltips', '#covid', '#ehealth', '#equitable',
    '#medstudents', '#medstudenttwitter', '#mhealth', '#pharmaceutical', '#coronavirus', '#covid19',
    '#genderaffirmingcare', '#nurses', '#telemedicine', '#transgendercare', '#vaccine', '#doctor',
    '#healthcareit', '#medtech', '#flu', '#hiv', '#pandemic', '#vetcare', '#aids', '#immunization',
    '#influenza'
]

Sec_retweet_hash = [
    '#lockdown', '#mentalhealth', '#quarantine', '#stayhome', '#vaccination', '#vaccines', '#who',
    '#disparities', '#healthit', '#hipaa', '#hospital', '#medicareforall', '#medtechvision22',
    '#pharmacies', '#publichealth', '#premisehealth', '#cancer', '#cancerawareness',
    '#healthcareprofessionals', '#hospitals', '#medicaid', '#patientexperience', '#qualitypatientcare',
    '#telehealthishealth', '#therapy', '#cardiology', '#nystateofhealth', '#patient', '#patientdata',
    '#patientengagement'
]

Thir_retweet_hash = [
    '#patients', '#physicians', '#psychiatry', '#psychology', '#telemed', '#wellness', '#breastcancer',
    '#drugdiscovery', '#heartfailure', '#medicaldeserts', '#utmhealthcare', '#aiforhealth',
    '#anesthesia', '#diagnostic', '#ruralhealth', '#soteradigitalhealth', '#vitalsignmonitoring',
    '#vitalsigns', '#weltelhealth', '#eyecare', '#eyedisease', '#healthcarecommercialintelligence',
    '#healthcareisahumanright', '#healthoutcomes', '#healthsecurity', '#hivcare', '#rehospitalizations',
    '#sarscov2', '#scrubs', '#selfcare'
]
```

Fig 3.6 Categorize HashTags

4. Visualizations

The following are the visualizations that we presented based on the data transformations that we performed in 3.1 .

The following graph types were used:

1. For distributions/counts of appointments for stand-alone categorical variables outlined in table 3.2, we used either pie charts or horizontal bar charts.(4.11, 4.16, 4.21). Pseudocode available in Appendix 2.1

2. For distributions/counts of appointments of the combination of categorical variables (Sec 3.1.a), we used stacked vertical bar charts & heatmaps showing both percentages and counts.(Fig 4.1 – 4.3, 4.5-4.6, 4.8, 4.10-4.11, 4.17-4.20, 4.22,4.24,4.25). Pseudocode available in Appendix 2.2

3. For distributions/counts of appointments of the combination of categorical variables involving time (Sec 3.1.a), we used dual-axis line and scatter plot showing both percentages and counts on different axis (Fig 4.4, 4.7, 4.9, 4.14, 4.15). Pseudocode available in Appendix 2.3

4. For distribution/counts of sentiments across relevant hashtags we used a heatmap.(Fig 4.26 & 4.27). Pseudo code in Appendix 3.1 & 3.2

4.1.a Visualizations Graphs

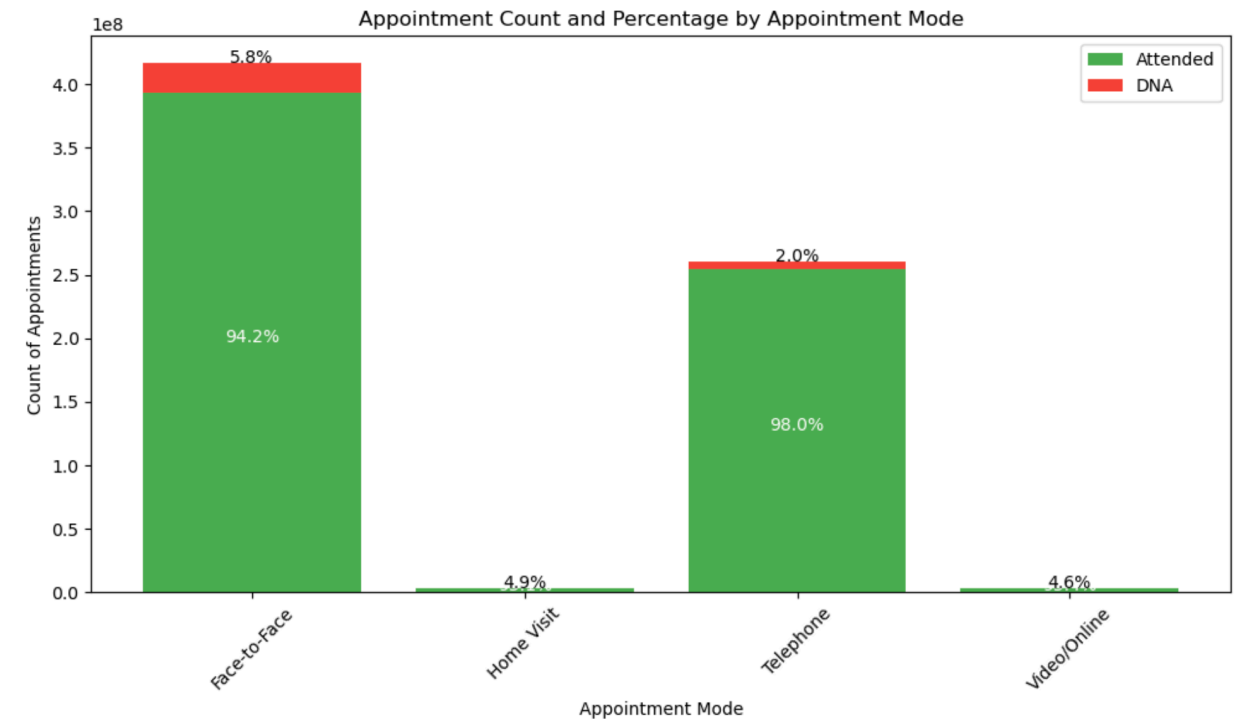


Fig 4.1

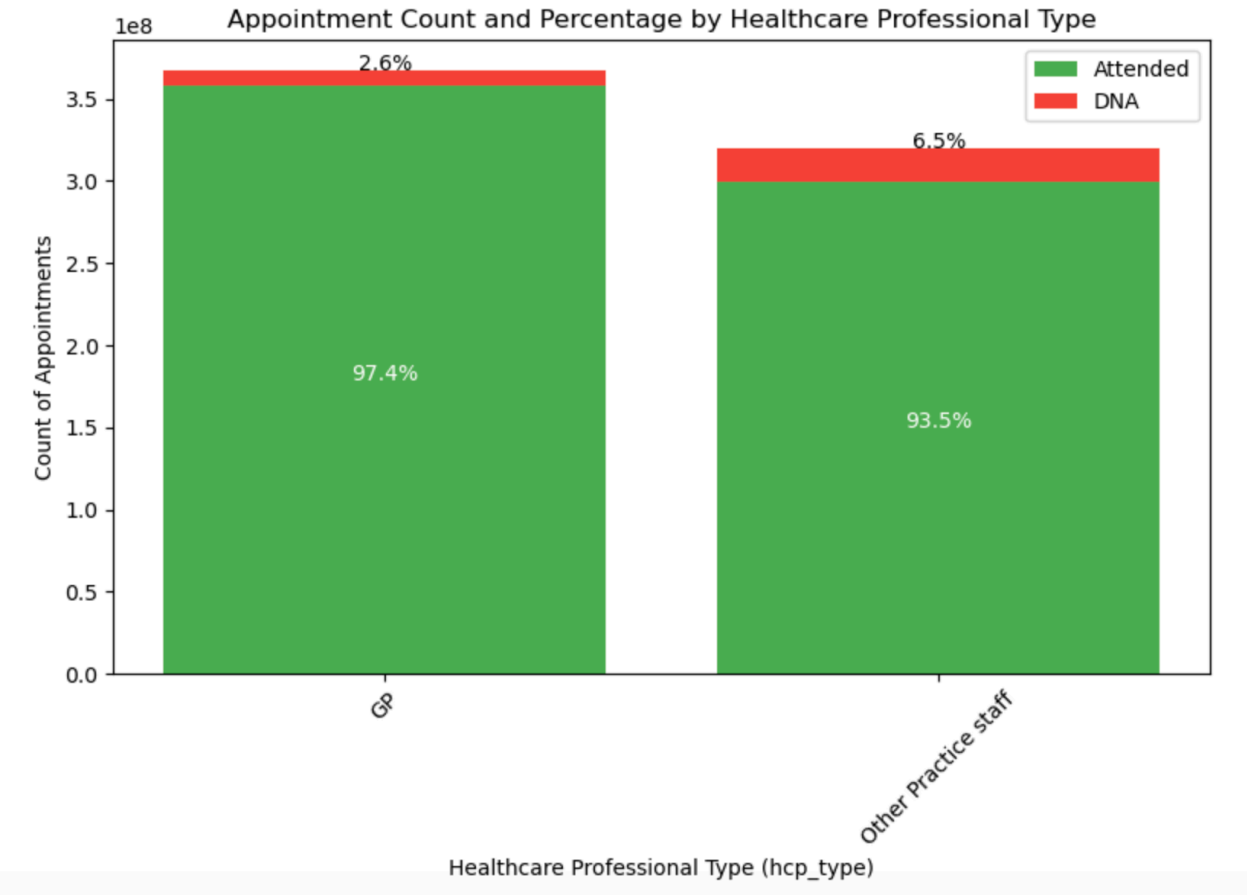

Appointment Count and Percentage by Healthcare Professional Type
1e8
Attended
DNA
2.6%
6.5%
97.4%
93.5%
Count of Appointments
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
GP
Other Practice staff
Healthcare Professional Type (hcp_type)

Fig 4.2

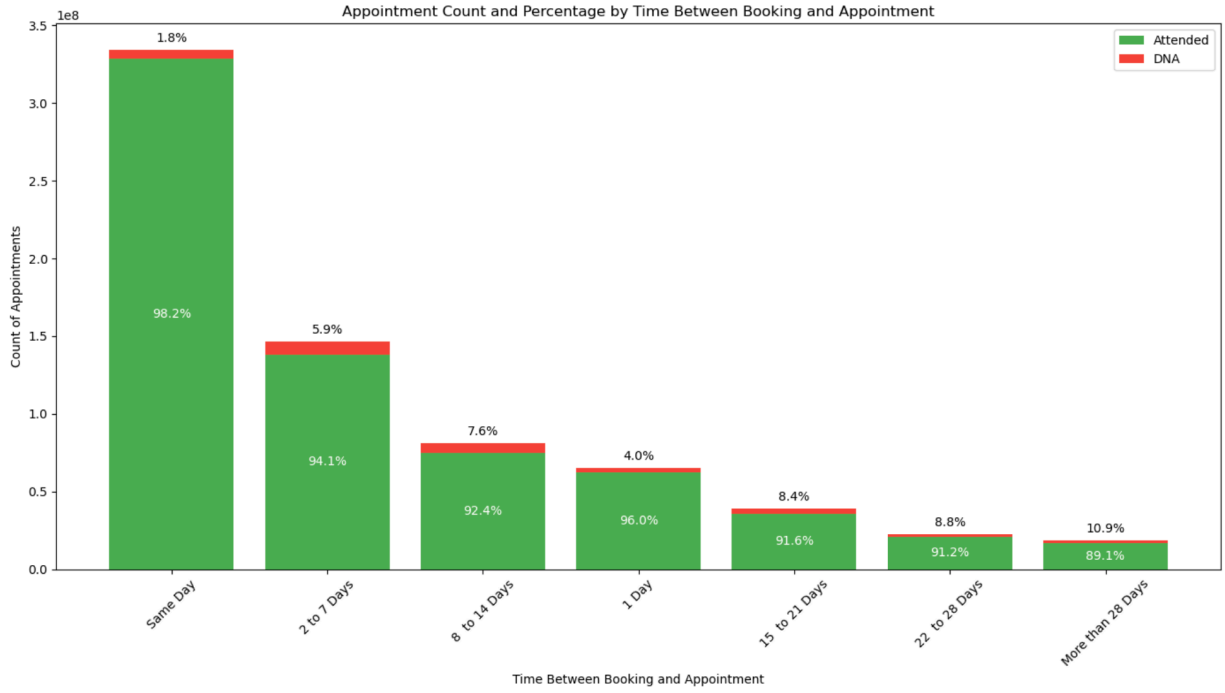

Appointment Count and Percentage by Time Between Booking and Appointment
1e8
Attended
DNA
1.8%
98.2%
5.9%
94.1%
7.6%
92.4%
4.0%
96.0%
8.4%
91.6%
8.8%
91.2%
10.9%
89.1%
Count of Appointments
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Same Day
2 to 7 Days
8 to 14 Days
1 Day
15 to 21 Days
22 to 28 Days
More than 28 Days
Time Between Booking and Appointment

Fig 4.3

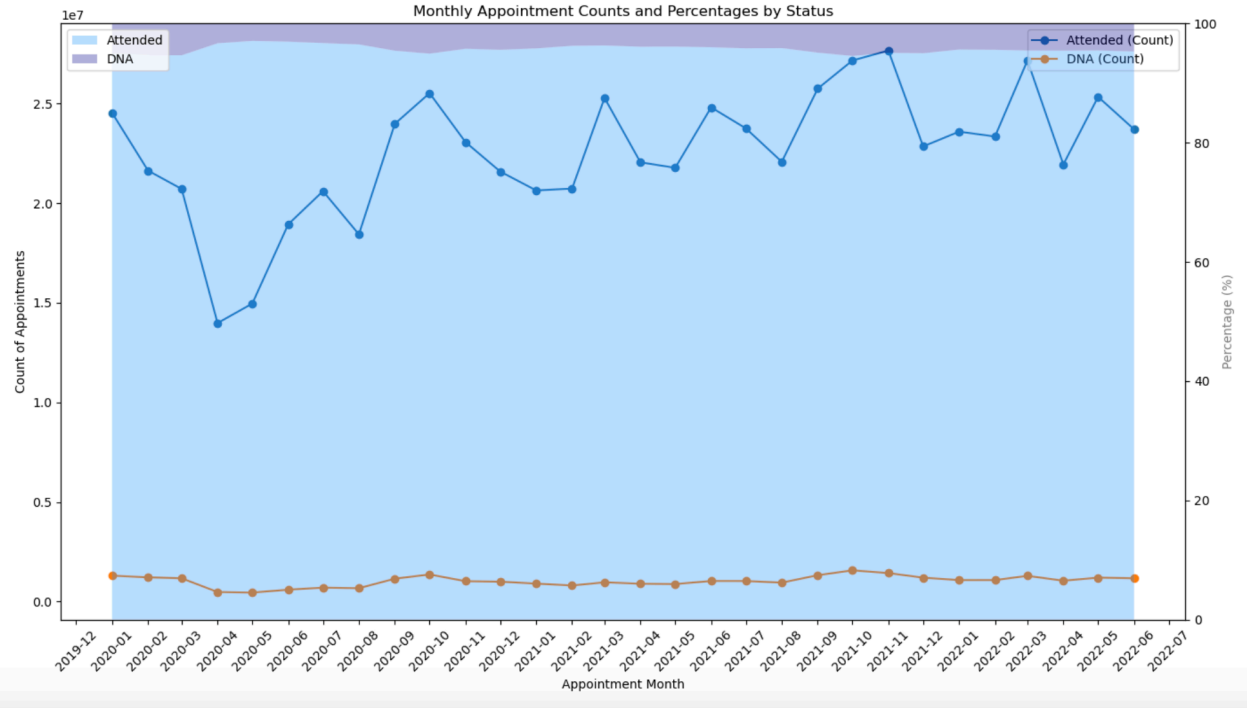

Monthly Appointment Counts and Percentages by Status
1e7
Attended
DNA
Attended (Count)
DNA (Count)
Count of Appointments
Percentage (%)
Appointment Month

Fig 4.4

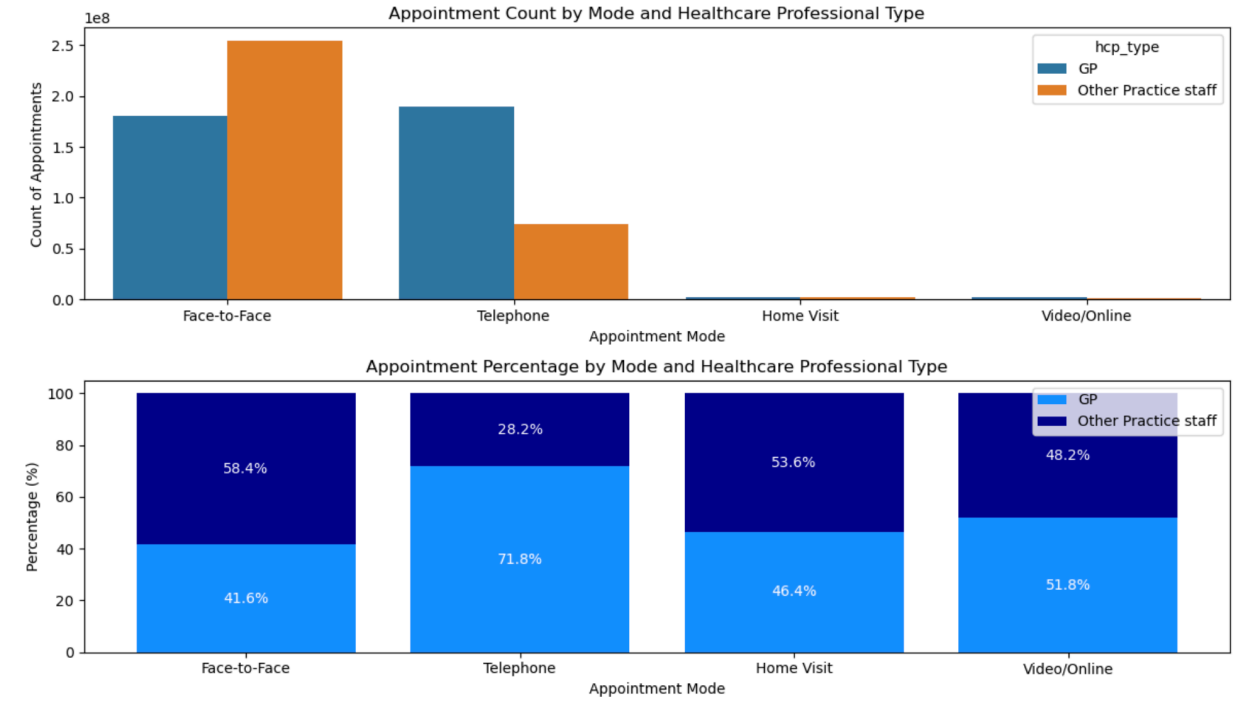

Appointment Count by Mode and Healthcare Professional Type
1e8
hcp_type
GP
Other Practice staff
Count of Appointments
Face-to-Face
Telephone
Home Visit
Video/Online
Appointment Mode
Appointment Percentage by Mode and Healthcare Professional Type
Percentage (%)
58.4%
41.6%
28.2%
71.8%
53.6%
46.4%
48.2%
51.8%
Appointment Mode

Fig 4.5

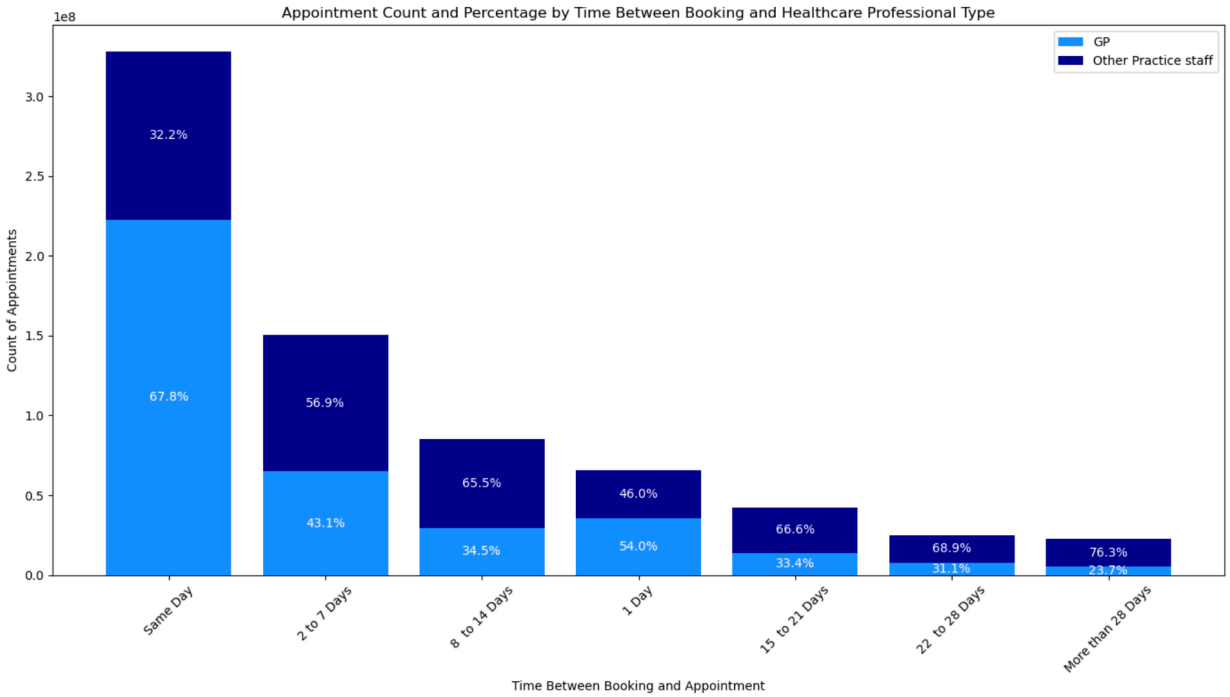

Appointment Count and Percentage by Time Between Booking and Healthcare Professional Type
1e8
GP
Other Practice staff
Count of Appointments
32.2%
67.8%
56.9%
43.1%
65.5%
34.5%
46.0%
54.0%
66.6%
33.4%
68.9%
31.1%
76.3%
23.7%
Same Day
2 to 7 Days
8 to 14 Days
1 Day
15 to 21 Days
22 to 28 Days
More than 28 Days
Time Between Booking and Appointment

Fig 4.6

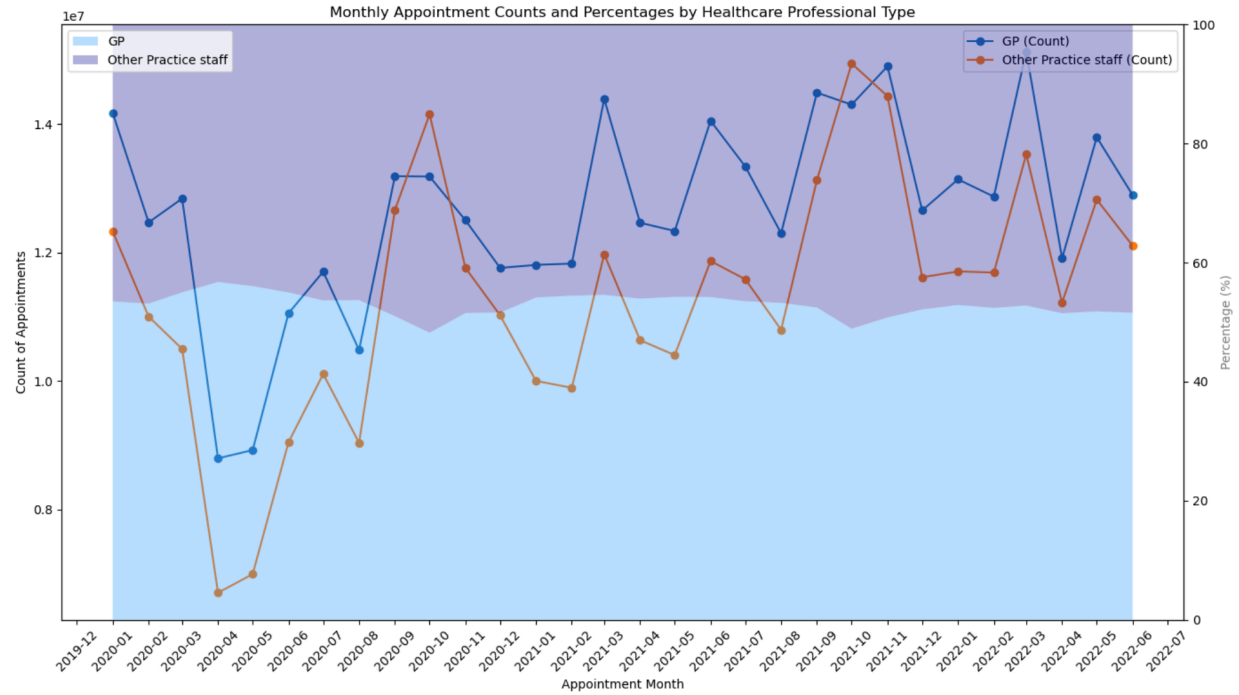

Monthly Appointment Counts and Percentages by Healthcare Professional Type
1e7
GP
Other Practice staff
GP (Count)
Other Practice staff (Count)
Count of Appointments
Percentage (%)
Appointment Month

Fig 4.7

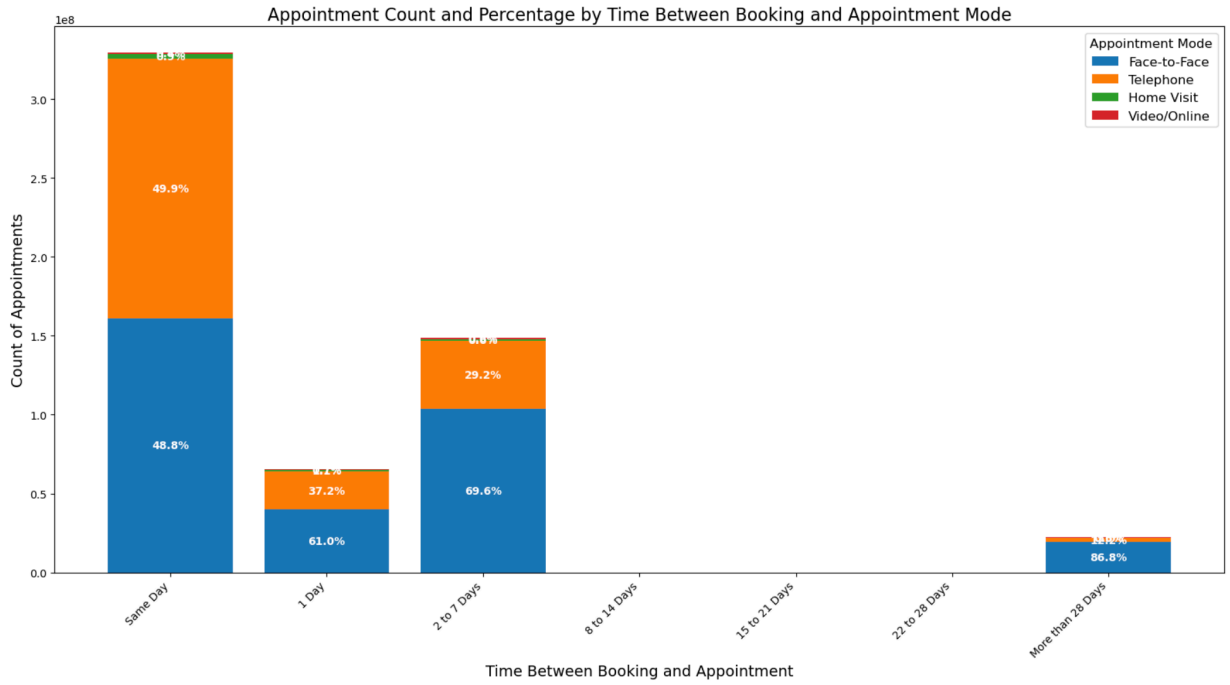

Appointment Count and Percentage by Time Between Booking and Appointment Mode
1e8
Appointment Mode
Face-to-Face
Telephone
Home Visit
Video/Online
Count of Appointments
49.9%
48.8%
37.2%
61.0%
29.2%
69.6%
86.8%
Same Day
1 Day
2 to 7 Days
8 to 14 Days
15 to 21 Days
22 to 28 Days
More than 28 Days
Time Between Booking and Appointment

Fig 4.8

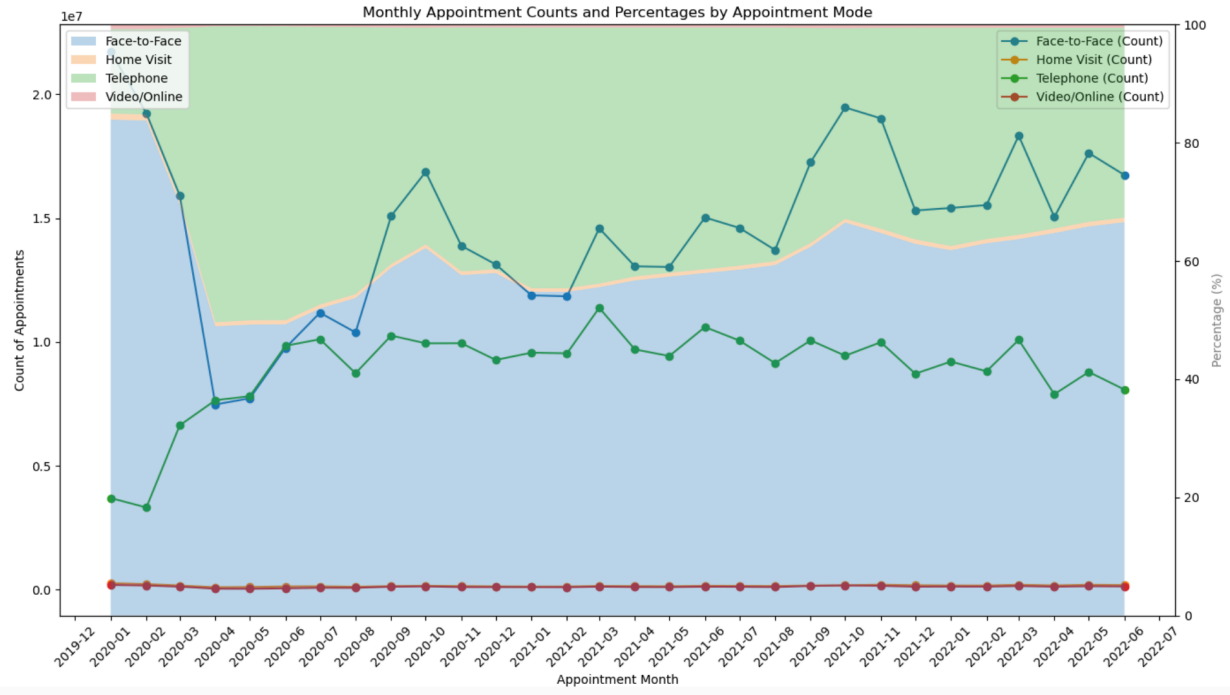

Monthly Appointment Counts and Percentages by Appointment Mode
1e7
Face-to-Face
Home Visit
Telephone
Video/Online
Face-to-Face (Count)
Home Visit (Count)
Telephone (Count)
Video/Online (Count)
Count of Appointments
Percentage (%)
Appointment Month

Fig 4.9

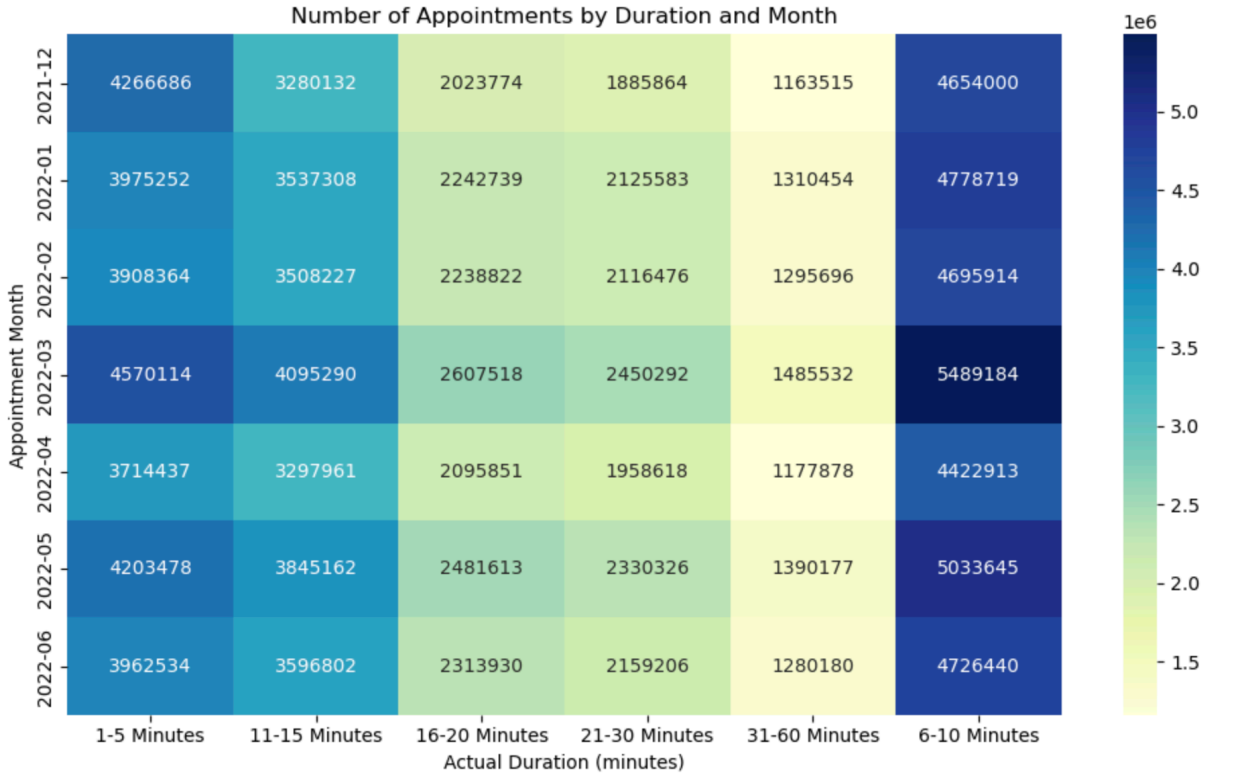


Fig 4.10

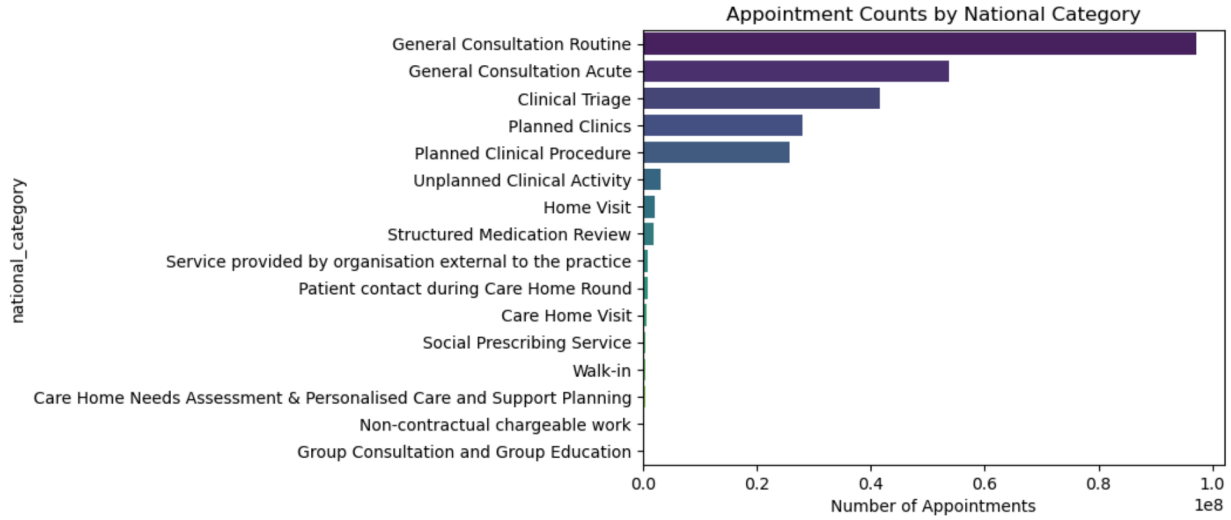


Fig 4.11

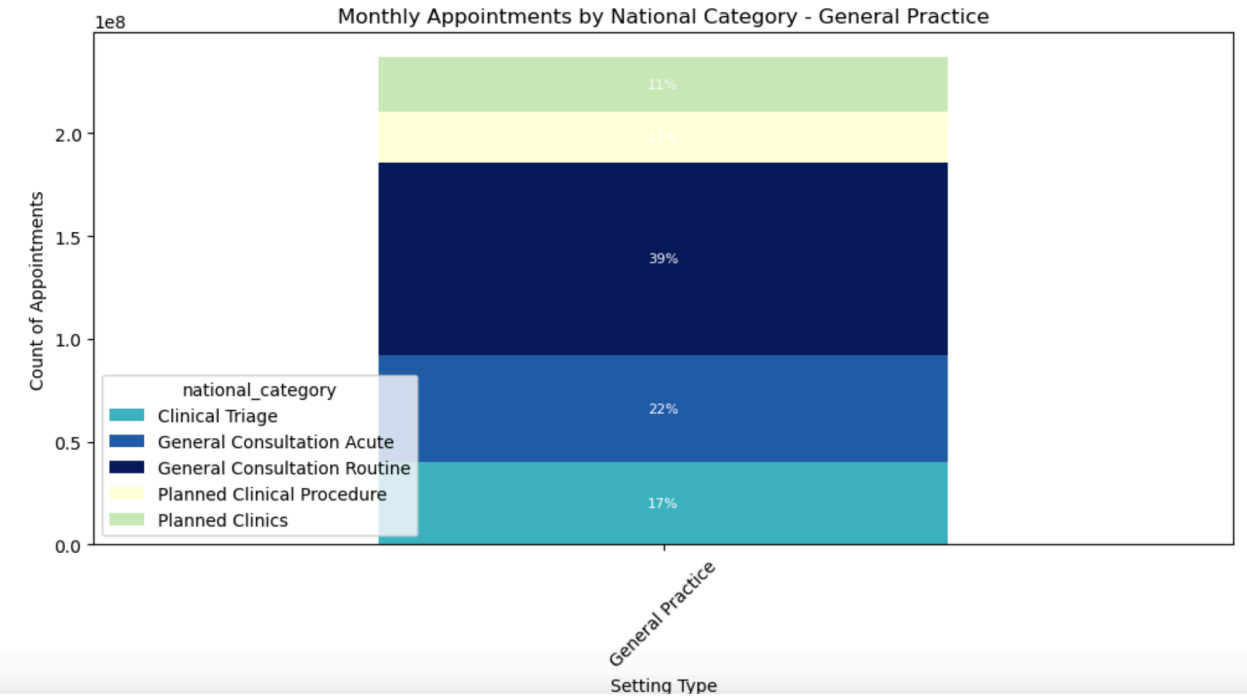


Fig 4.12

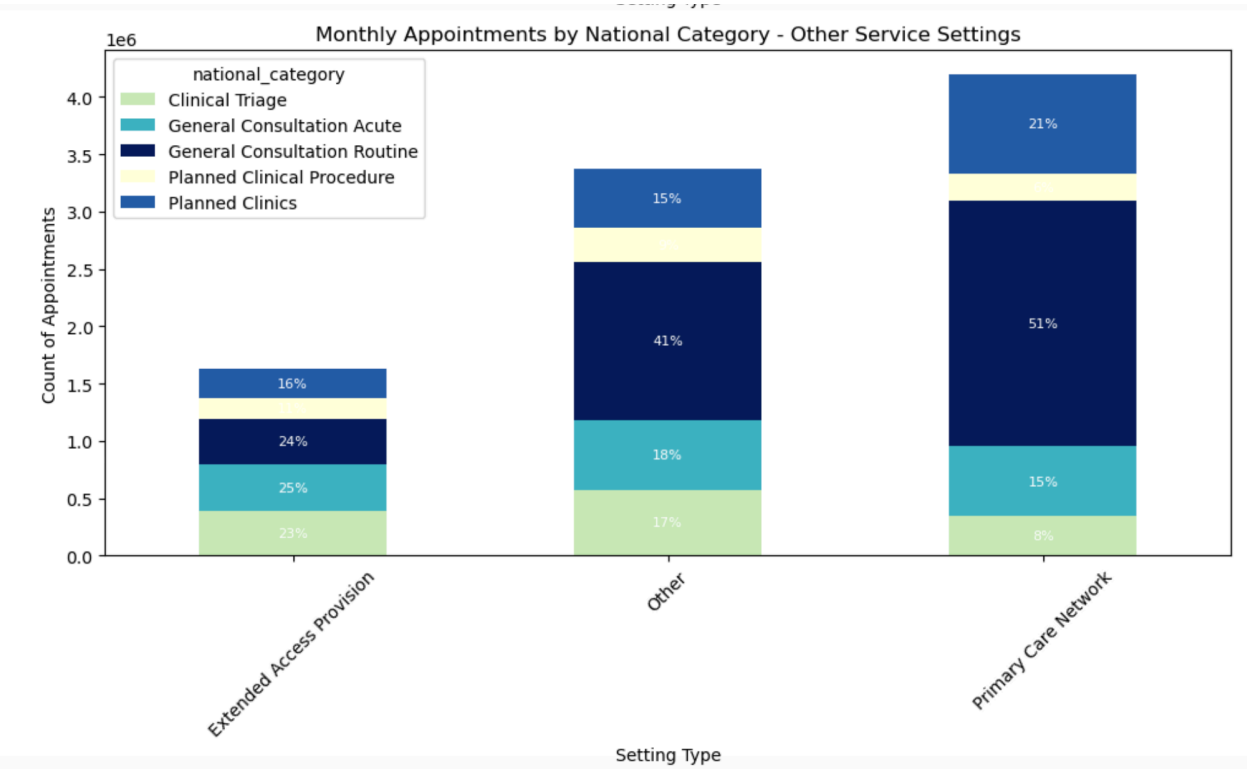


Fig 4.13

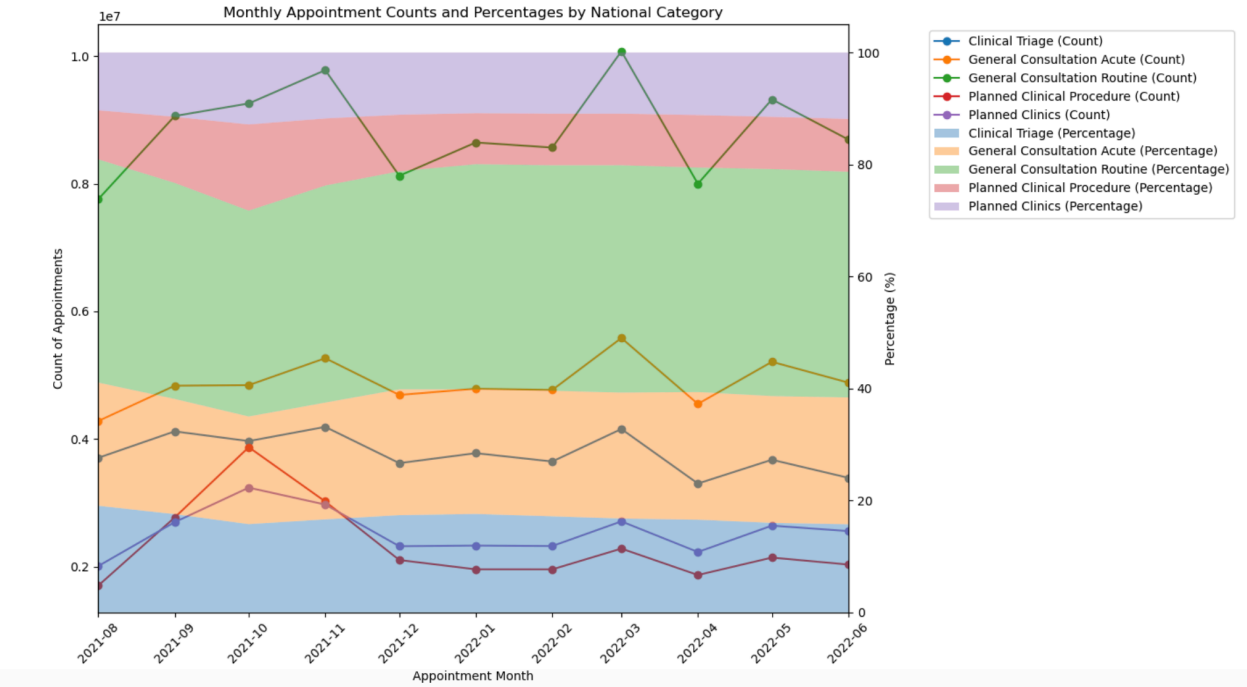


Fig 4.14

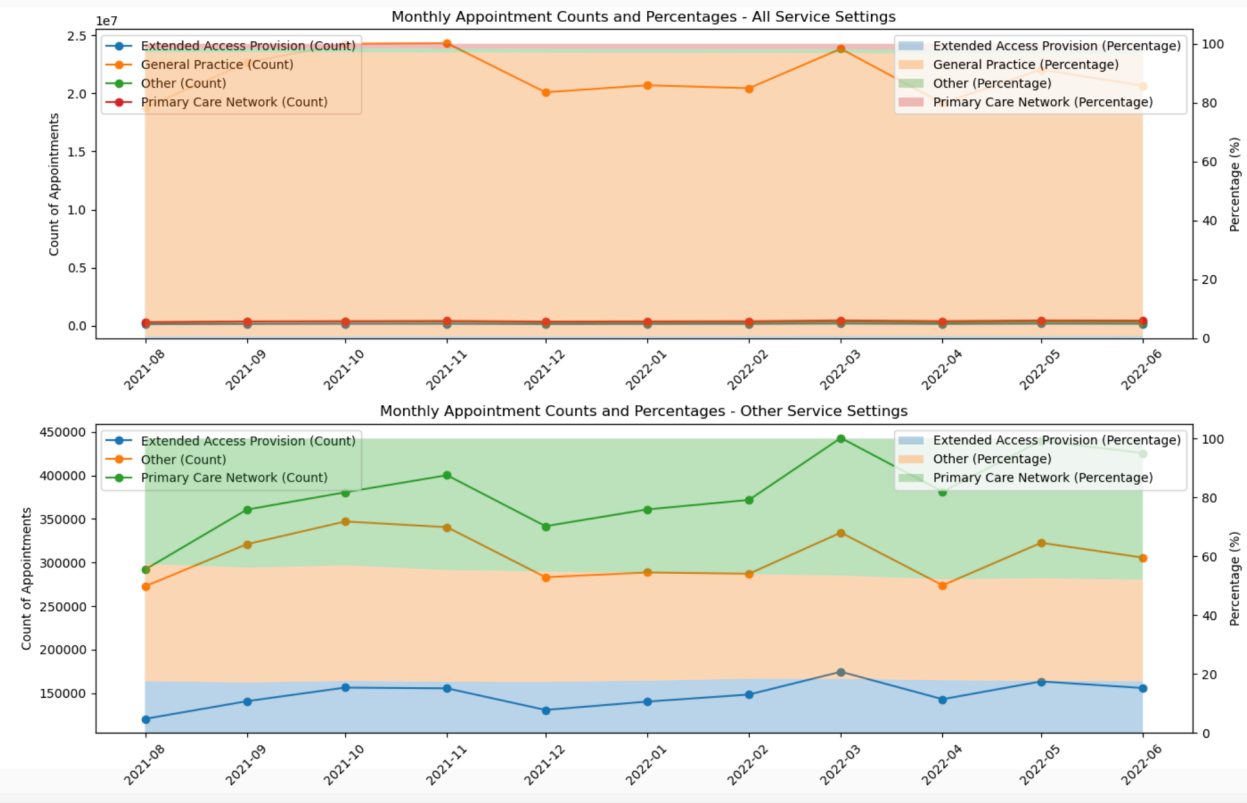


Fig 4.15

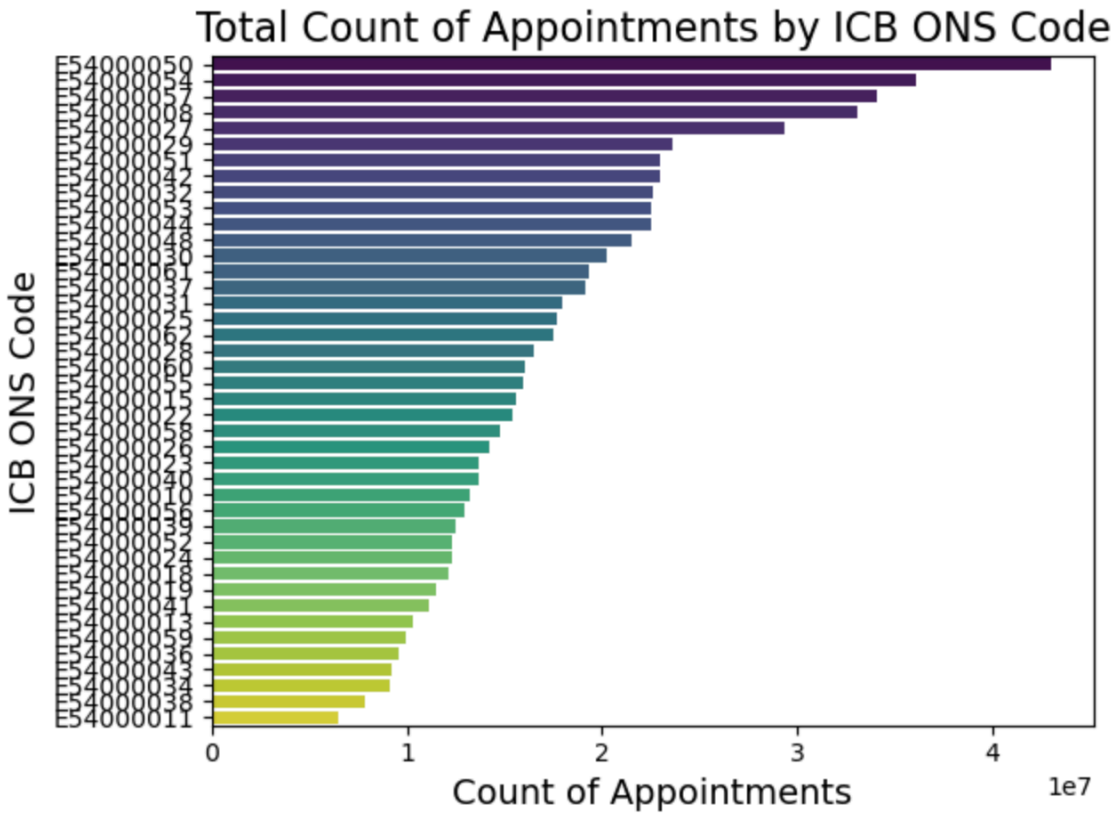


Fig 4.16

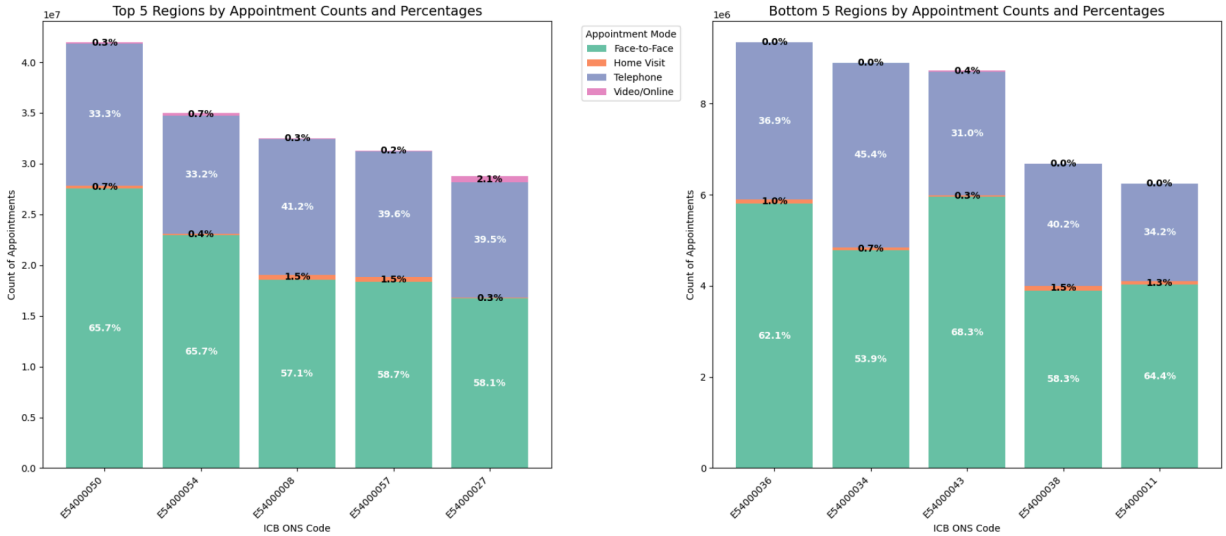


Fig 4.17

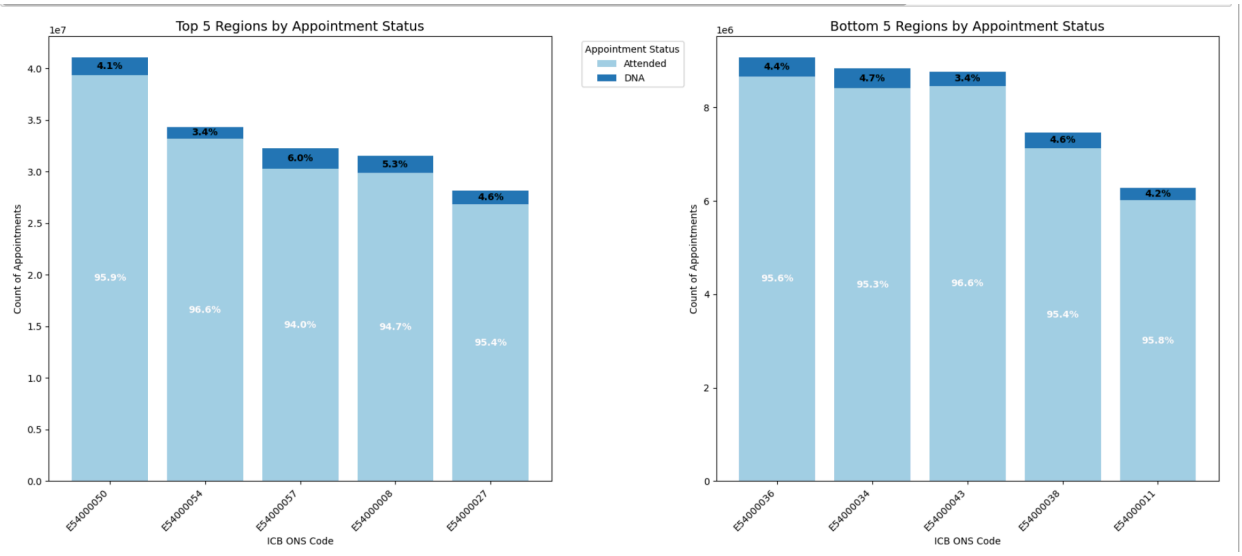


Fig 4.18

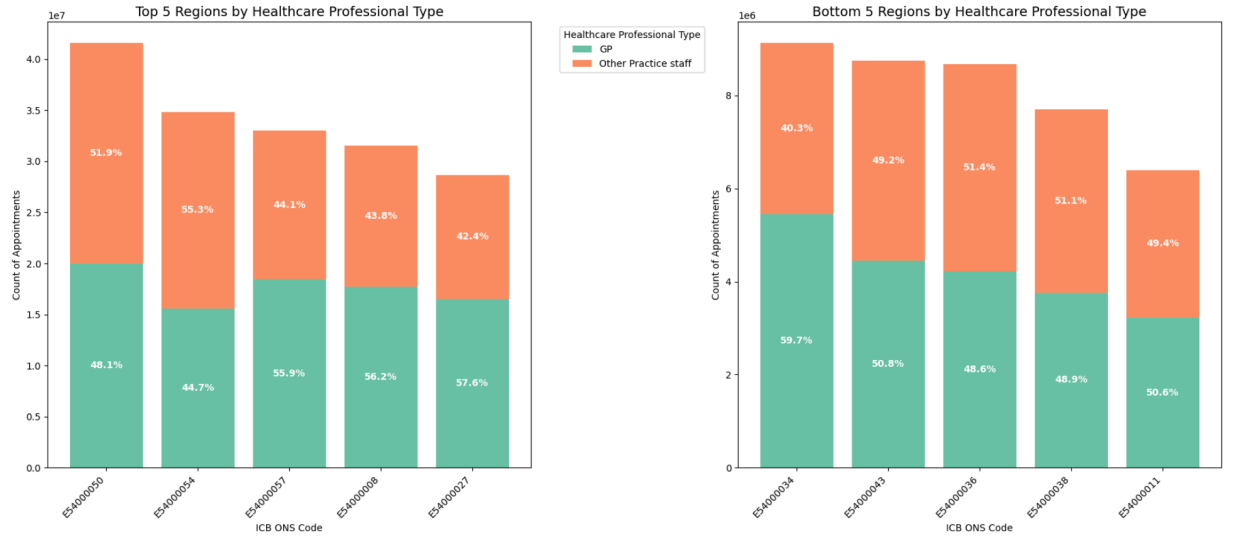


Fig 4.19

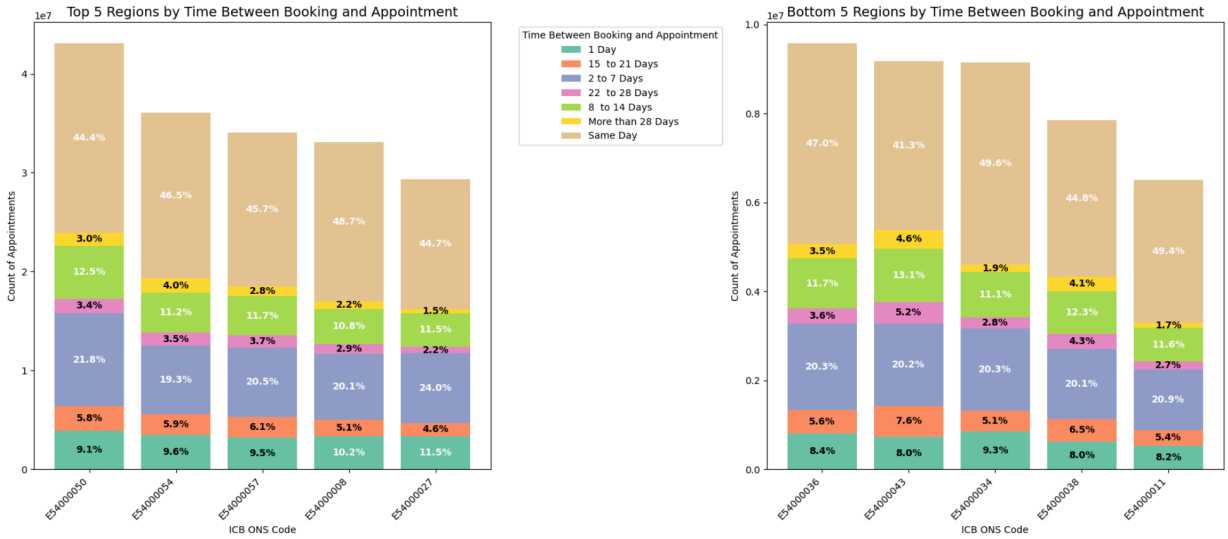

Top 5 Regions by Time Between Booking and Appointment
Bottom 5 Regions by Time Between Booking and Appointment
Time Between Booking and Appointment
1 Day
15 to 21 Days
2 to 7 Days
22 to 28 Days
8 to 14 Days
More than 28 Days
Same Day
Count of Appointments
ICB ONS Code
E54000050
E54000054
E54000057
E54000008
E54000027
E54000036
E54000043
E54000034
E54000038
E54000011

Fig 4.20

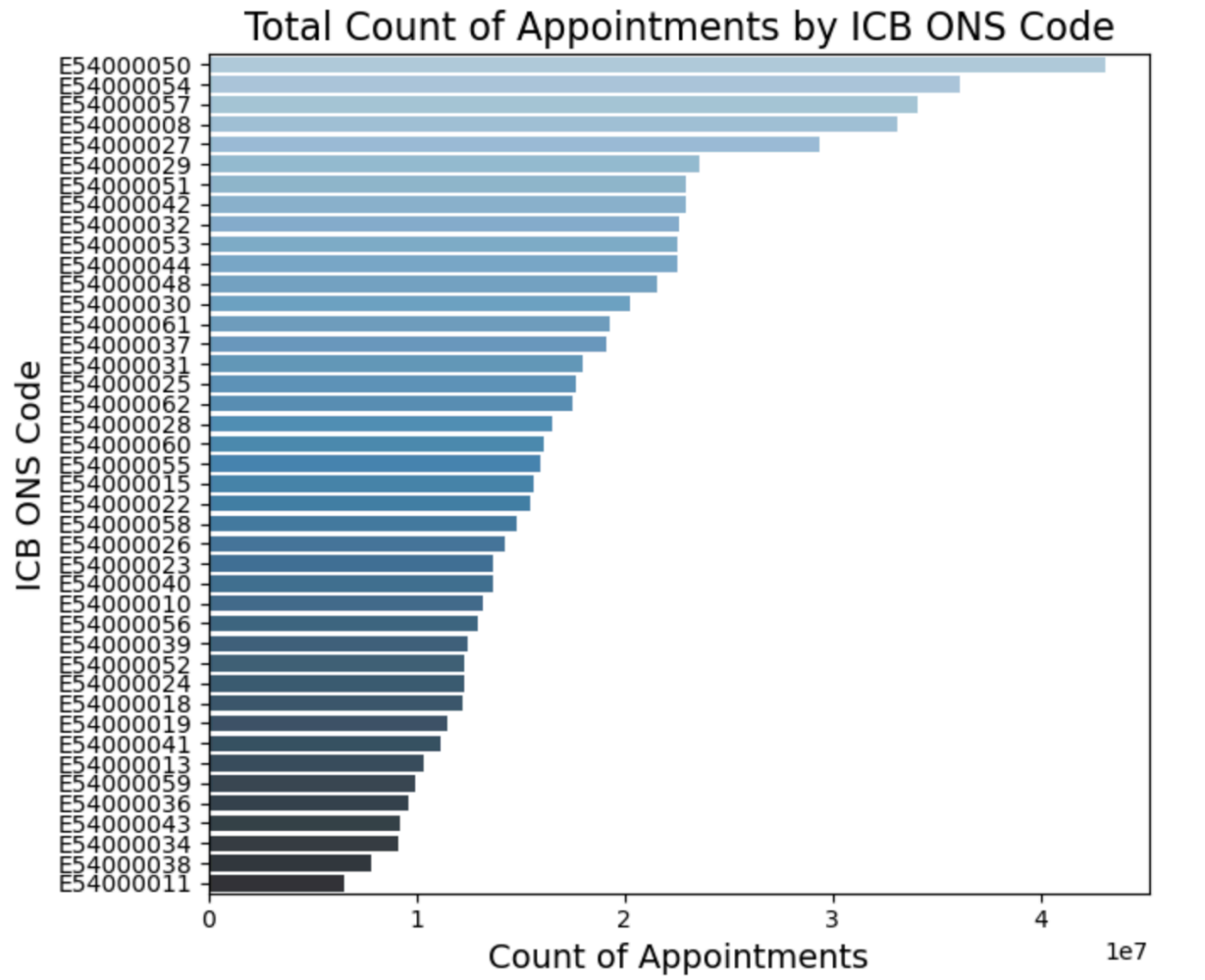

Total Count of Appointments by ICB ONS Code
ICB ONS Code
Count of Appointments
1e7

Fig 4.21

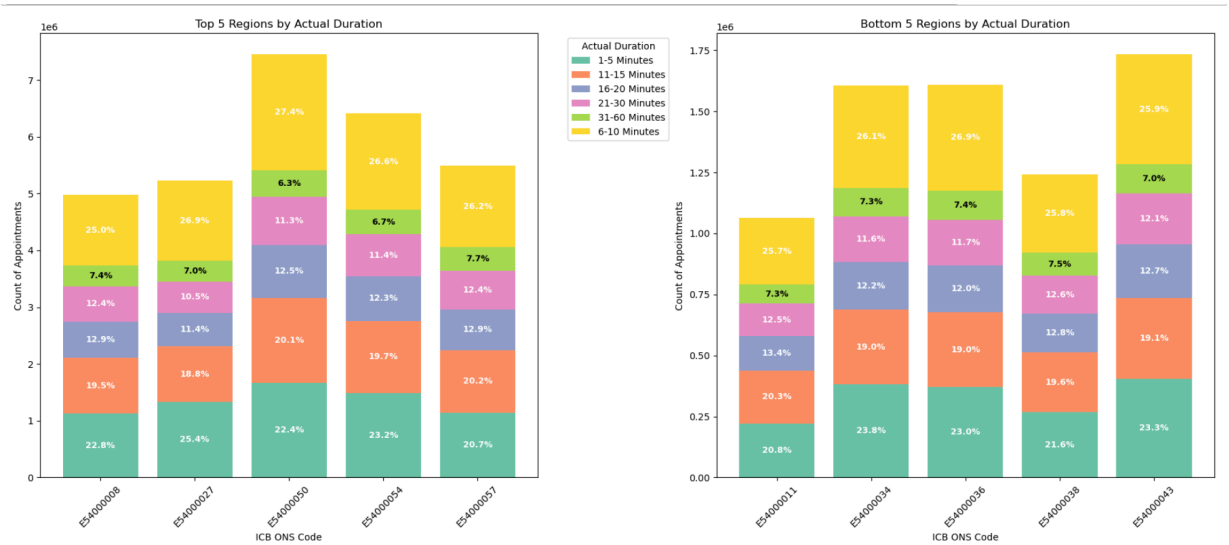


Fig 4.22

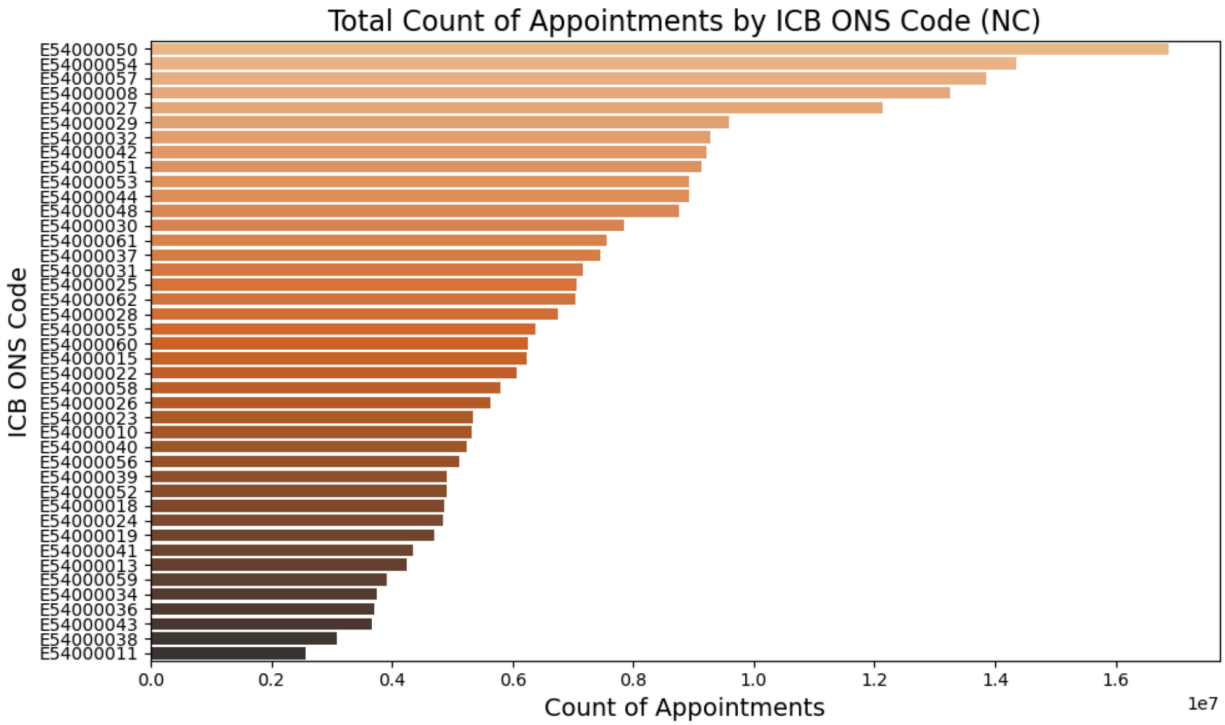


Fig 4.23



Fig 4.24

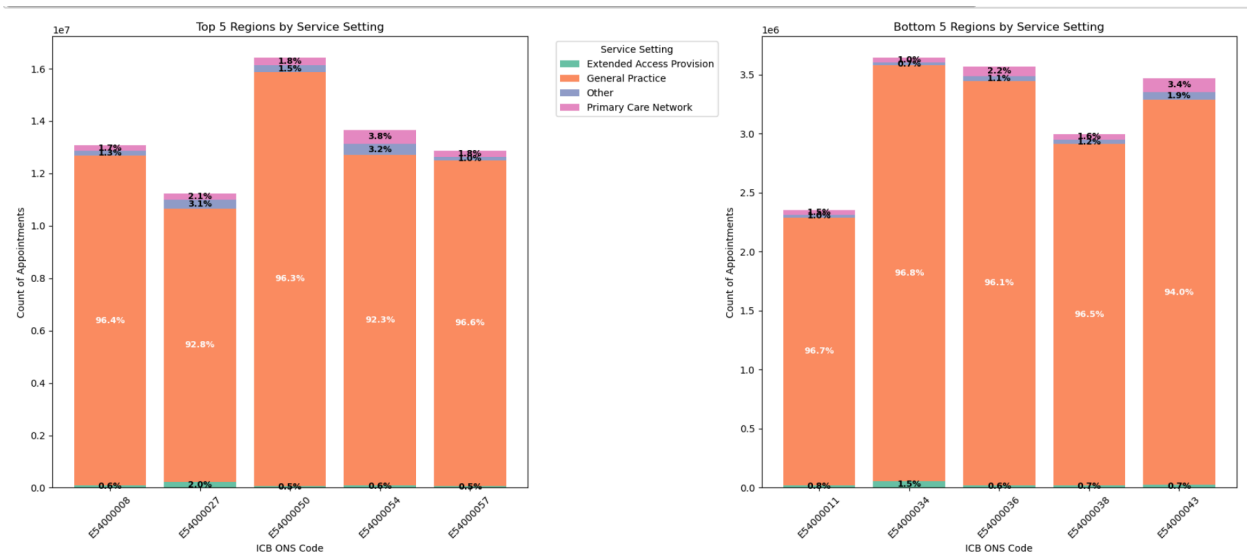


Fig 4.25

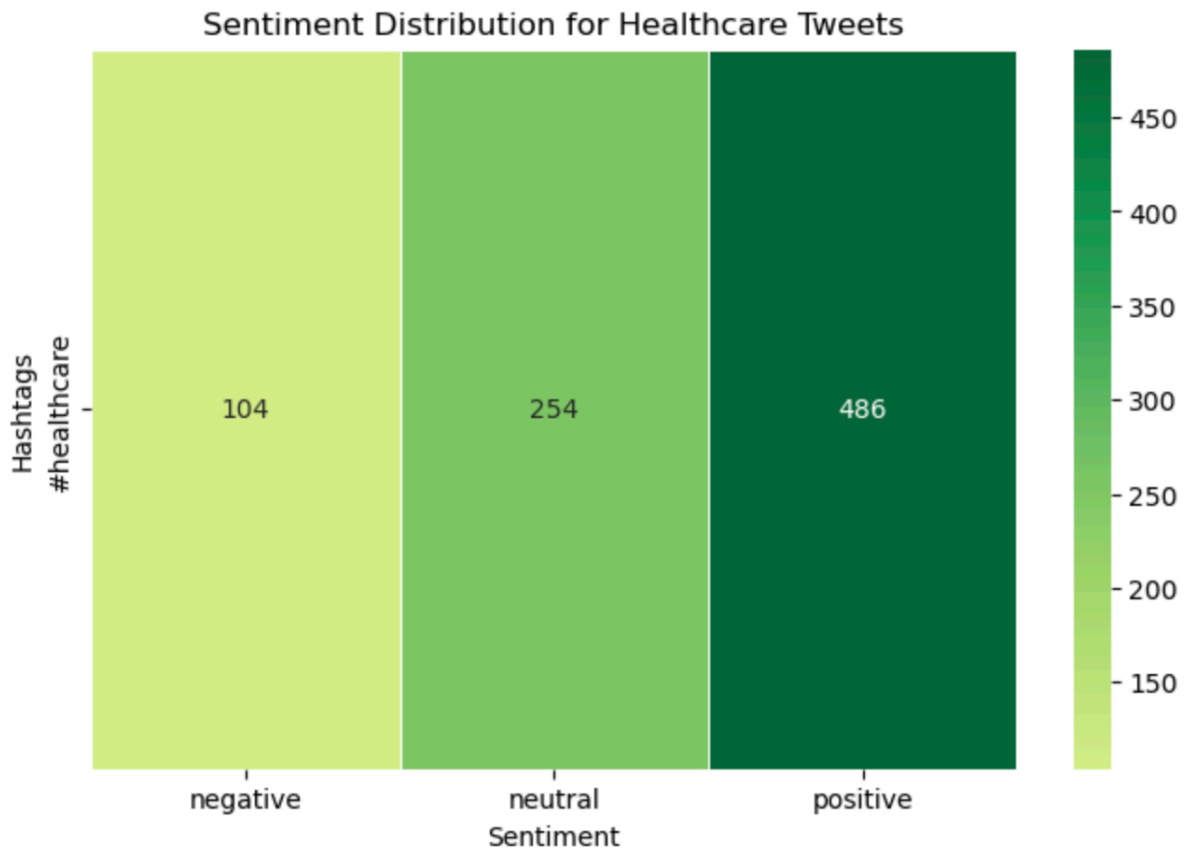


Fig 4.26

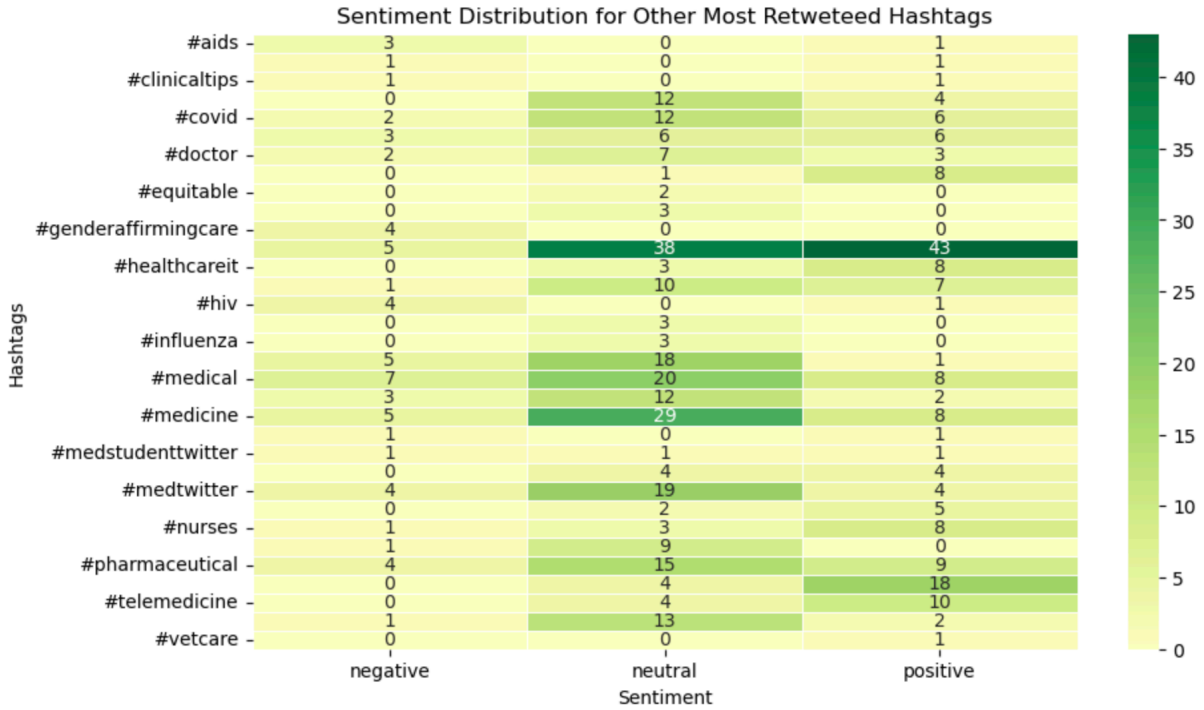


Fig 4.27

5. Observations & Recommendations

Our observations and recommendations are collated as per each dataset given in Table 5.1 to 5.3.

5.1.a High-Level Summary of Observations & Recommendations from Data Sets

Appointments Observations

Observation	Details
High and Consistent Attendance	Rates between 94.0% and 96.6%, with DNA below 6%.
GP Dominance	General Practitioners manage 43.8%–56.2% of appointments, representing 92% to 97% within general practice.
Primary Appointment Modes	Over 50% face-to-face and up to 41.2% telephone-based; home visits and video/online remain below 2.1%.
Short-Duration Appointments	1-10 minute slots make up over 46% of bookings, indicating a focus on brief consultations.
Booking Time Affects Attendance	Same-day bookings are 20.5%–21.8% with high attendance, while DNA rates increase for appointments over 28 days ahead.
Balanced Staffing with Regional Variations	GP shares range from 48.6% to 59.7%, with some regions relying more on other staff.
Underutilized Service Settings	Primary Care Networks and Extended Access Provision handle a small fraction of total appointments, presenting expansion opportunities.
Routine Consultations Dominant	Account for 40-45% of monthly appointments, whereas planned services like Clinical Procedures are underused.

Table 5.1

Recommendations

Recommendation	Details
Enhance Communication and Reminders	Improve patient communication channels and implement regular reminders to reduce missed appointments.
Expand Remote Options	Increase availability of telephone and online (video) appointments to provide greater flexibility and reduce DNAs.
Optimize GP and Staff Utilization	Assign more responsibilities to GPs and allocate other staff to handle future and non-urgent bookings.
Increase Urgent Care Capacity	Allocate additional resources to effectively manage same-day face-to-face and telephone appointments.
Leverage Primary Care Networks	Redirect routine and planned appointments to Primary Care Networks and Extended Access Provision to alleviate GP workload.
Adjust Staffing for Demand Peaks	Temporarily increase staffing and appointment slots during high-demand periods and peak months.
Support Community and Elderly Care	Enhance Care Home Needs Assessments and Home Visits to meet the demands of community and elderly care.

Table 5.2

Twitter Recommendations

Recommendation	Details
Enhance Patient Communication	Use hashtags like **#healthcare** and **#health** to share important information, improving understanding and attendance.
Promote Digital Health Services	Expand telehealth options by leveraging positive sentiments toward **#telehealth**, **#healthtech**, and **#digitalhealth**.
Increase Awareness of Specialized Services	Boost engagement with **#mentalhealth** and **#ruralhealth** by raising awareness and accessibility of these services.
Leverage Public Interest in Health Topics	Utilize enthusiasm for **#wellness**, **#selfcare**, **#vaccination**, and **#publichealth** to promote wellness programs and health initiatives.

Table 5.3

Detailed version of observations & recommendations are found in Appendix 1

6. Appendix

Appendix 1

1. Appointments Related observations & recommendations

A. Appointments Regional

Table	Categorical Variable 1	Categorical Variable 2	Key Observations	Recommendations
Appointments Regional	Appointment Status	Appointment Mode	- **Face-to-Face Dominant**: Majority are face-to-face appointments (94.2% attended, 5.8% DNA), showing a preference or need for in-person visits. - **High Telephone Attendance**: Telephone appointments have the highest attendance (98% attended, 2% DNA), indicating convenience boosts attendance. - **Low Use of Home Visits and Video/Online**: These modes are minimal but have high attendance rates (Home Visits 95.1%, Video/Online 95.4%).	- **Reduce Face-to-Face DNAs**: Implement improved communication channels with patients and prompt rescheduling to decrease missed face-to-face appointments. - **Expand Telephone options**: Offer more telephone channels to enhance access and reduce DNAs. - **Promote Online Modes**: Encourage Video/Online appointments for patients facing barriers to in-person visits.
		Healthcare Professional Type	- **High GP Attendance**: GP appointments have a 97.4% attendance rate (2.6% DNA), showing strong commitment. - **Higher DNA for Other Staff**: Appointments with other staff have 93.5% attendance (6.5% DNA).	- **Expand GP Services**: Consider giving GPs more responsibilities to boost engagement and attendance. - **Reinforce Reminders for Other Staff**: Use extra reminders or education to emphasize attending these appointments.
		Time Between Booking and Appointment	- **Higher Attendance with Shorter Booking**: Same-day bookings have the highest attendance (98.2% attended, 1.8% DNA). - **DNA Increases with Longer Booking**: DNA rates rise with longer intervals, up to 10.9% for bookings over 28 days. - **Attendance Drops After Two Weeks**: Attendance falls below 92% for bookings beyond 14 days.	- **Encourage Shorter Bookings**: Offer shorter booking intervals, especially for patients prone to DNAs. - **Implement Reminders for Long-Leads**: Send frequent reminders as the appointment nears to reduce DNAs. - **Investigate Long-Lead DNAs**: Understand why patients miss long-lead appointments to improve interventions.
		Appointment Month	- **Stable Attendance, Low DNAs**: Most appointments are attended monthly, showing strong adherence. - **Slight Variations Over Time**: Appointment counts fluctuate slightly, possibly due to seasons or external events. - **COVID-19 Impact**: Notable dip in early to mid-2020 aligns with the pandemic's onset.	- **Maintain Regular Communication**: Keep patient reminders consistent to sustain high attendance. - **Monitor External Factors**: Anticipate dips by tracking events that affect attendance. - **Analyze Pandemic Impact**: Study how COVID-19 affected appointments to prepare for future disruptions.
	Healthcare Professional Type	Appointment Mode	- **GPs Lead Telephone Appointments**: GPs handle most telephone appointments; other staff slightly more in face-to-face mode. - **Appointment Mode Preferences**: Reflects role specialization and patient preferences.	- **Expand GP Telephone Consultations**: Increase to optimize resources and reduce wait times. - **Optimize Staff Scheduling**: Assign other staff to future bookings, freeing GPs for immediate needs. - **Promote Remote Options**: Encourage Video/Online appointments where appropriate.
		Time Between Booking and Appointment	- **GPs Handle Immediate Appointments**: 67.8% of same-day appointments are with GPs. - **Staff Shift Over Booking Time**: Other staff handle more delayed appointments, up to 76.3% for bookings over 28 days.	- **Increase GP Availability for Urgent Cases**: Allocate more GP slots for urgent appointments. - **Optimize Staff for Future Appointments**: Schedule other staff for future bookings to maximize efficiency.
		Appointment Month	- **Monthly Appointment Fluctuations**: Variations suggest staffing adjustments or changing demand. - **Possible Staffing Adjustments**: Fluctuations may reflect responses to seasonal demand or external factors like holidays or pandemics.	- **Optimize Resources Based on Trends**: Anticipate high-demand periods to ensure adequate staffing. - **Adjust Staffing**: Adapt to changes in demand to maintain consistent patient access.
	Appointment Mode	Time Between Booking and Appointment	- **Same-Day Mainly Face-to-Face and Telephone**: 48.8% Face-to-Face, 49.9% Telephone for immediate care. - **Shift in Mode Over Time**: Face-to-Face reliance grows with longer bookings, up to 69.6% for 2-7 days ahead. - **Minimal Use of Home Visit and Video/Online**: Low percentages across all booking times.	- **Enhance Capacity for Urgent Care**: Allocate resources to meet same-day Face-to-Face and Telephone demand. - **Adjust Services Based on Booking Patterns**: Strategically allocate resources for expected demand. - **Expand Remote Options**: Promote Video/Online and Home Visits to reduce pressure on other modes.
		Appointment Month	- **Dominance of Face-to-Face and Telephone Modes**: Face-to-Face and Telephone appointments consistently make up the majority, with Telephone appointments peaking early 2020 due to COVID-19 before stabilizing. - **Increasing Home Visits**: Home Visits have slightly increased since mid-2021 as pandemic restrictions eased. - **Low Video/Online Usage**: Video/Online appointments remain minimal, indicating limited telemedicine adoption despite the pandemic.	- **Expand Remote Care Options**: Increase Video/Online appointments to enhance accessibility for non-urgent consultations. - **Allocate Resources for In-Person and Telephone Modes**: Ensure sufficient staffing and resources for high-demand Face-to-Face and Telephone appointments. - **Monitor Home Visit Trends**: Track rising demand for Home Visits to address in-home care needs.

Table 5.1

B. Actual Duration

Table	Categorical Variable/Level 1	Categorical Variable 2	Key Observations	Recommendations
Actual Duration	**Appointment Duration**: 1-15 Minutes	**Appointment Month**: All Months	- Consistently high number of short-duration appointments across all months. - Indicates significant workload involves brief consultations or check-ins.	- **Optimize Short-Duration Appointments**: Automate these interactions where possible (e.g., online check-ins, virtual follow-ups) to reduce staff burden and increase efficiency.
	Appointment Duration: 16-30 Minutes	**Appointment Month**: May & March 2022 (Peak), All Months	- High and consistent number of appointments within the 16-30 minute range. - Peaks in May & March 2022. - Likely represents standard consultations, a core component of scheduling.	- **Adjust Duration-Based Capacity**: Optimize scheduling and staffing for these in-demand durations to reduce wait times and improve patient satisfaction. - **Allocate More Resources** During Peak Months: Anticipate demand peaks and adjust staffing and resources accordingly to manage increased workload effectively.
	Appointment Duration: 31-60 Minutes	**Appointment Month**: All Months	- Relatively low but stable number of longer appointments. - These are less frequent and possibly reserved for complex or detailed consultations.	- **Monitor Longer Appointments**: Ensure scheduling systems can accommodate these adequately to avoid bottlenecks, especially if demand for complex cases increases.

Table 5.2

C. National Category

Table	Variable 1	Categorical Variable 2	Key Observations	Recommendations
National Category	National Category	Service Setting	**General Practice Dominance**: High volume in General Practice, handling most appointments—93.4 million routine (39%) and 52 million acute (22%). Planned services like Clinical Procedures and Clinics are underused, each making up only 11% despite totaling 49 million appointments.	- **Redistribute Workload**: Encourage use of Primary Care Networks and Extended Access Provision to reduce General Practice's load. - **Expand Planned Services**: Promote Planned Procedures and Clinics to manage appointments more efficiently.
			Underutilization of Other Service Settings: Primary Care Networks manage 4.1 million appointments, mainly routine consultations (51%). Extended Access Provision is underused with only 1.5 million appointments. Other settings have smaller volumes and are far behind General Practice.	- **Leverage Primary Care Networks**: Increase their role in Planned Procedures and Acute Care. - **Expand Extended Access Provision**: Shift routine appointments from General Practice to these settings.
		Appointment Month	**Dominance of Routine Consultations**: Routine consultations account for 40-45% of appointments monthly, with stable demand. Planned categories like Clinical Procedures and Clinics are underused, staying below 15% of appointments.	- **Expand Routine Capacity**: Allocate resources to meet high demand. - **Promote Planned Services**: Raise awareness to redistribute some load from routine consultations.
			Seasonal Variations: Minor peaks in acute consultations during certain months, but overall stable distribution of appointment types over time.	- **Adjust for Seasonal Demand**: Temporarily increase slots or staff during peak times. - **Explore Alternatives**: Use telehealth or streamlined pathways during busy periods.
	Service Setting	Appointment Month	**Underutilization of Extended Access Provision**: Low usage with around 100,000 appointments monthly (0.5-1% of total), indicating potential capacity, especially for planned and routine services.	- **Increase Utilization**: Promote this setting to distribute patient load and ease General Practice burden. - **Raise Awareness**: Educate patients about this option.
			Growth in Non-Core Settings: Non-core settings like Primary Care Networks are slightly increasing from 1% to 3% of appointments, but counts fluctuate.	- **Support Non-Core Settings**: Enhance capacity and encourage use to relieve General Practice. - **Encourage Diversification**: Implement incentives and policies to promote these services.
			Fluctuations in Other Settings: Variable demand in non-core settings could lead to bottlenecks without flexible resource allocation.	- **Flexible Resources**: Manage demand peaks with dynamic staffing or scheduling. - **Prevent Bottlenecks**: Monitor trends to adjust resources proactively.

Table 5.3

Geographical Related Information

A. Appointments Regional

Table	Categorical Variable 1	Categorical Variable 2	Key Observations	Recommendations
Appointments Regional	Appointment Status	Top 5 Regions	- **High Attendance Rate**: Attendance rates range from **94.0% to 96.6%**. - **Low DNA Rates**: DNA rates are below **6%**, with the highest in E54000057 at **6.0%**.	- **Maintain High Attendance**: Continue effective communication and reminder systems. - **Targeted Interventions**: For regions like E54000057 with higher DNA rates, implement additional reminders or flexible scheduling.
		Bottom 5 Regions	- **Consistent Attendance Rates**: Ranging from **95.3% to 96.6%**. - **Low DNA Variation**: DNA rates range from **3.4% to 4.7%**.	- **Monitor Trends**: Regularly monitor attendance to address any emerging issues. - **Sustain Effective Practices**: Continue current strategies to maintain high attendance rates.
	Healthcare Professional Type	Top 5 Regions	- **GP Dominance**: General Practitioners handle **43.8% to 56.2%** of appointments. - **Higher GP Representation**: Regions like E54000057 have GP representation at **55.9%**.	- **Utilize Other Staff**: Increase use of Other Practice staff to reduce GP workload. - **Balance Staffing**: Encourage strategies for balanced staff utilization.
		Bottom 5 Regions	- **Balanced Staffing**: GP shares range from **48.6% to 59.7%**. - **Variation in Staff Utilization**: Regions like E54000034 rely more on Other Staff (**59.7%**).	- **Leverage Best Practices**: Use balanced staffing models as a template for other regions. - **Optimize Resources**: Adjust staffing based on regional demands to enhance patient satisfaction.
	Time Between Booking and Appointment	Top 5 Regions	- **High Same-Day Appointments**: **20.5% to 21.8%** are same-day bookings. - **Short-Term Planning**: Substantial bookings within **1-7 days**. - **Notable Longer Lead Times**: Up to **48.7%** of appointments have a **15-21 days** lead time.	- **Increase Immediate Care Capacity**: Add resources for same-day appointments. - **Adjust Appointment Allocation**: Balance appointment slots to meet urgent care demand and patient preferences.
		Bottom 5 Regions	- **Longer Lead Times Dominance**: Up to **49.4%** of appointments have a **15-21 days** lead time. - **Lower Same-Day Demand**: Same-day bookings as low as **8.4%**. - **Consistent 1-7 Days Bookings**: Short-term bookings are present but less dominant.	- **Encourage Advance Booking**: Promote initiatives for patients to book earlier. - **Optimize Scheduling**: Adjust appointment slots to improve resource utilization and accommodate patient needs.
	Appointment Mode	Top 5 Regions	- **Face-to-Face Mode Dominance**: Face-to-Face appointments range from **57.1% to 65.7%**. - **Telephone Usage**: Significant telephone appointments, e.g., **41.2%** in E54000057 and **39.6%** in E54000008. - **Minimal Video/Online Usage**: Peaks at **2.1%** in E54000027.	- **Increase Digital Access**: Expand digital options in regions with high Face-to-Face usage. - **Optimize Face-to-Face Resources**: Ensure sufficient staffing and facilities; implement scheduling flexibility to manage peak hours.
		Bottom 5 Regions	- **Face-to-Face Mode Prevalence**: Dominant at **53.9% to 68.3%**. - **Lower Telephone Usage**: Some regions rely heavily on Face-to-Face, e.g., E54000036 at **62%**. - **Home Visits and Video/Online Usage**: Negligible, with Home Visits around **1%** and Video/Online **0.4%** or less.	- **Promote Remote Options**: Encourage use of Telephone and Video/Online appointments to alleviate pressure on in-person facilities. - **Optimize Resource Allocation**: Adjust resources to support increased remote appointment uptake.

Fig 5.4

B. Actual Duration

Table	Categorical Variable 1	Categorical Variable 2	Key Observations	Recommendations
Appointment Duration	Appointment Duration	Top 5 Regions	- **6-10 Minute Appointments Dominate**: Constituting **26.2% to 27.4%** of appointments. - **Considerable Short Durations (1-5 minutes)**: Over **20%** in regions like E54000050 and E54000054. - **Lower 16-20 Minute Appointments**: Only **6-7%** across regions.	- **Analyze Duration Patterns**: Understand services within common timeframes to optimize scheduling. - **Improve Efficiency for Short Appointments**: Streamline processes to handle more cases. - **Benchmark Across Regions**: Identify best practices to improve service quality. - **Strategic Resource Allocation**: Focus on common durations while preparing for longer ones.
		Bottom 5 Regions	- **6-10 Minute Duration Dominance**: Ranging from **25.7% to 26.9%**. - **Higher 1-5 Minute Appointments**: Values from **20.8% to 23.8%**, higher than top regions. - **Minimal 31-60 Minute Appointments**: Indicates preference for shorter times.	- **Optimize Scheduling**: Adjust to accommodate higher shares of short appointments. - **Resource Alignment**: Ensure resources match duration patterns to enhance efficiency. - **Apply Similar Strategies**: Tailor recommendations to the specific needs of bottom regions.

Fig 5.5

C. National Category

Table	Categorical Variable 1	Categorical Variable 2	Key Observations	Recommendations
National Category	National Category	Top 5 Regions	- **High Appointment Concentrations**: Notable clusters in **General Consultation Routine** and **General Consultation Acute**, with regions like **E54000050** and **E54000054** recording over **6 million** appointments. - **High Clinical Triage Counts**: Over **2 million** appointments in top regions. - **More Care Home and Home Visits**: Indicates greater emphasis on community and elderly care.	- **Enhance Service Capacity in Bottom Regions**: Boost availability of high-demand services like **General Consultation Acute** and **Clinical Triage**. - **Focus on Community and Elderly Care**: Increase **Care Home Needs Assessments** and **Home Visits** in bottom regions. - **Targeted Investment in Structured Reviews**: Invest in resources for **Structured Medication Reviews** to streamline care. - **Regional Service Optimization**: Tailor services based on regional demographics and health needs.
		Bottom 5 Regions	- **Lower Counts in High-Demand Services**: Fewer appointments in **General Consultation Acute** and **Clinical Triage**. - **Less Emphasis on Community Care**: Fewer **Care Home Needs Assessments** and **Home Visits**, suggesting regional differences in service emphasis or population needs. - **Consistent Lower Demand Services**: **Structured Medication Review** and **Unplanned Clinical Activities** are lower but stable.	- **Balance Service Accessibility**: Enhance availability of high-demand services to match top regions. - **Address Underserved Needs**: Focus on increasing community and elderly care services. - **Efficiently Handle Structured Reviews**: Allocate resources to manage consistent demand in lower-demand services. - **Support Service Expansion**: Assist bottom regions in enhancing offerings where demand is rising.
	Service Setting	Top 5 Regions	- **Dominance of General Practice**: Accounts for **92% to 97%** of appointments. - **Higher Diversity in Services**: Slightly higher percentages in **Extended Access Provision** and **Primary Care Network**, with up to **3.8%** in specific ICB codes. - **Low Representation of Other Categories**: Minimal but slightly more prominent than in bottom regions.	- **Strengthen Extended Access**: Expand **Extended Access Provision** and **Primary Care Network** options. - **Promote Alternative Service Offerings**: Explore **Primary Care Network** initiatives or community-based care to relieve pressure on General Practice. - **Optimize Resource Allocation**: Support alternative settings to balance healthcare resource distribution.
		Bottom 5 Regions	- **Slightly Higher General Practice Percentage**: Even more reliant on General Practice than top regions. - **Lower Diversity in Services**: Lower percentages in **Extended Access Provision** and **Primary Care Network**. - **Minimal Other Categories**: Very low representation of other service settings.	- **Expand Service Diversity**: Introduce more **Extended Access Provision** and **Primary Care Network** services. - **Focus on Alternative Settings**: Promote community-based care to enhance service delivery. - **Balance Resource Distribution**: Allocate resources to alternative settings where demand is increasing.

Fig 5.6

Twitter Related Information

Tweet Hashtags	Observations:	Recommendations:
Highest Retweeted	1. **Prominent Neutral Sentiment in Healthcare Discussions** - Hashtags like **#healthcare** and **#health** have high volumes of positive & neutral sentiment tweets. - Indicates that healthcare-related tweets are often informational or descriptive.	1. **Leverage Neutral Sentiment for Informative Outreach** - **Enhance Patient Education**: Use the positive & neutral tone to disseminate important healthcare information, supporting efforts to maintain high attendance rates and encourage advance booking. - **Standardize Communication**: Provide clear updates on services like appointment modes and durations, aligning with recommendations on optimizing resource utilization.
2nd Highest Retweeted	2. **Positive Sentiment Linked with Telehealth and Health Technology** - Hashtags such as **#telehealth**, **#healthtech**, and **#digitalhealth** show higher positive sentiments. - Reflects favorable public perception of technological advancements.	2. **Promote Digital Health Initiatives** - **Expand Telehealth Services**: Capitalize on positive sentiment by promoting telehealth options, supporting recommendations to increase digital access and remote appointment modes. - **Educate on Technology Benefits**: Highlight convenience and efficiency of digital services, aiding in reducing strain on face-to-face appointments.
3rd Highest Retweeted	3. **Limited Engagement with Specialized and Sensitive Topics** - Hashtags like **#genderaffirmingcare**, **#mentalhealth**, and **#ruralhealth** have fewer mentions but varied sentiments.	3. **Targeted Communication for Underserved Services** - **Raise Awareness of Specialized Services**: Inform patients about services in areas like mental health and gender-affirming care, complementing recommendations to diversify - **Enhance Accessibility in Rural Areas**: Focus on improving services in rural regions, supporting prior recommendations on addressing accessibility issues.
4th Highest Retweeted	4. **Absence of Strong Negative Sentiment** - Most healthcare-related hashtags lack significant negative sentiment.	4. **Proactive Public Engagement** - **Strengthen Positive Messaging**: Provide positive updates to maintain public perception, reinforcing efforts to promote preventive health. - **Address Concerns Promptly**: Monitor for emerging negative sentiments to address issues quickly, supporting overall service satisfaction.
	Additional Observations: 5. **Interest in Wellness and Self-Care Topics** - Hashtags like **#wellness** and **#selfcare** show predominance of positive sentiment. - Indicates public enthusiasm for wellness initiatives.	**Additional Recommendations**: 5. **Integrate Wellness Programs into Services** - **Expand Preventive Care Offerings**: Incorporate wellness programs into services, aligning with recommendations to promote preventive health. - **Leverage Social Media for Wellness Education**: Use positive sentiment to share wellness tips, enhancing patient education efforts.
	6. **Low Engagement with Healthcare Technology Terms** - Hashtags like **#medtech** and **#healthcaretechnology** have limited mentions. - Reflects a gap in public awareness of healthcare technology advancements.	6. **Enhance Public Awareness of Healthcare Technology** - **Simplify Messaging Around Technology**: Create accessible content explaining benefits of healthcare technology, supporting recommendations to improve public understanding. - **Demonstrate Technology's Impact**: Share success stories about telehealth to encourage utilization, aiding in optimizing resource allocation.
	7. **Engagement with Vaccination and Public Health Topics** - Hashtags **#vaccination** and **#publichealth** show high engagement.	7. **Leverage Public Interest for Health Campaigns** - **Promote Vaccination Programs**: Continue vaccine awareness campaigns, aligning with efforts to improve service adoption. - **Inform on Public Health Initiatives**: Educate the public on health measures, supporting recommendations to use engagement data for outreach.
	8. **Growing Focus on Mental Health** - Hashtag **#mentalhealth** shows sustained interest. - Highlights increasing recognition of mental health in healthcare.	8. **Expand Mental Health Services and Awareness** - **Integrate Mental Health into Primary Care**: Enhance mental health resources within general practice, supporting recommendations to diversify offerings. - **Public Education Campaigns**: Raise awareness about mental health services, complementing efforts to improve patient engagement

Fig 5.7

Appendix 2

2.1

B. Data Preprocessing & Transformation

Filtering Out Tweets Data with Missing Hashtags

```
# Filter out rows where the 'tweet_entities_hashtags' column has missing values (NaN)
# This ensures that only tweets with hashtags are included in the filtered DataFrame
tweets_fil = tweets[~tweets["tweet_entities_hashtags"].isna()]

# Print the shape of the filtered DataFrame 'tweets_fil' to check the number of rows and columns
# This helps verify how many tweets with hashtags remain after filtering
print(tweets_fil.shape)
```

```
(1007, 10)
```

2.2

```
# Import necessary libraries
import pandas as pd
import datetime

# Step 1: Split hashtags into a list
# Split the hashtags column into lists (assuming they are comma-separated)
tweets_fil['hashtags_list'] = tweets_fil['tweet_entities_hashtags'].str.split(',')

# Step 2: Expand hashtags to individual rows
# Explode the hashtags list so each hashtag appears in its own row
tweets_exploded = tweets_fil.explode('hashtags_list')

# Step 3: Standardize hashtags
# Convert hashtags to lowercase for consistency
tweets_exploded['hashtags_list_lower'] = tweets_exploded['hashtags_list'].str.lower()
# Trim any leading or trailing spaces from the hashtags
tweets_exploded['hashtags_list_lower'] = tweets_exploded['hashtags_list_lower'].str.strip()

# Step 4: Define function to extract date from tweet_id
def get_date_from_tweet_id(tweet_id):
    # Ensure the tweet_id is an integer
    tweet_id = int(tweet_id)
    # Twitter epoch in milliseconds
    twitter_epoch = 1288834974657
    # Shift the tweet_id to remove lower 22 bits and calculate the timestamp in ms
    timestamp_ms = (tweet_id >> 22) + twitter_epoch
    # Convert milliseconds to seconds
    timestamp_s = timestamp_ms / 1000.0
    # Create a datetime object
    tweet_datetime = datetime.datetime.utcfromtimestamp(timestamp_s)
    # Return only the date part
    return tweet_datetime.date()

# Step 5: Apply date extraction function to get posting dates
# Add a new column 'Date' with the posting date derived from tweet_id
tweets_exploded["Date"] = tweets_exploded['tweet_id'].apply(get_date_from_tweet_id)
```

2.3

Sentiment Analysis on Tweets Data

```
!pip install nltk
!pip install vaderSentiment
!pip install textblob
!pip install spacy
!pip install spacytextblob
!pip install transformers
!pip install torch
```

2.4

```
import nltk
# Download the VADER lexicon for sentiment analysis (only needed once)
nltk.download('vader_lexicon')

from nltk.sentiment.vader import SentimentIntensityAnalyzer

# Initialize the VADER sentiment analyzer
sia = SentimentIntensityAnalyzer()

# Define a function to get sentiment scores for a given text using VADER
def get_sentiment(text):
    # Use VADER to analyze the text and return a dictionary of sentiment scores
    # Scores include 'neg' (negative), 'neu' (neutral), 'pos' (positive), and 'compound' (overall sentiment)
    sentiment = sia.polarity_scores(text)
    return sentiment

import re

# Define a function to preprocess the tweet text for better sentiment analysis
def preprocess_text(text):
    # Remove URLs by identifying patterns starting with 'http' or 'www'
    text = re.sub(r'http\S+|www.\S+', '', text)
    # Remove mentions (e.g., @username) and hashtags (e.g., #hashtag)
    text = re.sub(r'@\w+|#\w+', '', text)
    # Optionally remove special characters and numbers, retaining only letters and whitespace
    text = re.sub(r'[^A-Za-z\s]', '', text)
    # Strip leading and trailing whitespace for a cleaner result
    text = text.strip()
    return text

# Apply text preprocessing to each tweet, storing cleaned text in a new column 'clean_text'
tweets_exploded['clean_text'] = tweets_exploded['tweet_full_text'].apply(preprocess_text)

# Apply the sentiment analysis function to each cleaned tweet
# Store the sentiment scores (neg, neu, pos, and compound) in a new column 'sentiment_scores'
tweets_exploded['sentiment_scores'] = tweets_exploded['clean_text'].apply(get_sentiment)
```

```
# Apply text preprocessing to each tweet, storing cleaned text in a new column 'clean_text'
tweets_exploded['clean_text'] = tweets_exploded['tweet_full_text'].apply(preprocess_text)

# Apply the sentiment analysis function to each cleaned tweet
# Store the sentiment scores (neg, neu, pos, and compound) in a new column 'sentiment_scores'
tweets_exploded['sentiment_scores'] = tweets_exploded['clean_text'].apply(get_sentiment)

# Extract the 'compound' score from the sentiment scores and store it in 'compound_score' column
# Compound score represents the overall sentiment ranging from -1 (most negative) to +1 (most positive)
tweets_exploded['compound_score'] = tweets_exploded['sentiment_scores'].apply(lambda x: x['compound'])

# Define a function to classify sentiment based on the compound score
def classify_sentiment(compound_score):
    # Classify as 'positive' if compound score >= 0.05
    if compound_score >= 0.05:
        return 'positive'
    # Classify as 'negative' if compound score <= -0.05
    elif compound_score <= -0.05:
        return 'negative'
    # Otherwise, classify as 'neutral'
    else:
        return 'neutral'

# Apply the classification function to each compound score
# Store the result in a new column 'sentiment' for easy sentiment analysis
tweets_exploded['sentiment'] = tweets_exploded['compound_score'].apply(classify_sentiment)
```

2.5

Keyword Extraction and Filtering for Tweets Data

```
import re
import pandas as pd

# Assuming you have a DataFrame 'tweets_exploded' with a column 'tweet_full_text'

# Your existing keyword list
keywords = [
    'nhs',
    'gp',
    'bma',
    'doctor',
    'medication',
    'appointment',
    'appointments',
    'healthcare',
    'health',
    'patient',
    'patients',
    'consultation',
    'medicare',
    'medical',


]
```

```python
# Step 1: Create a mapping of concatenated keywords to their spaced versions
multi_word_keywords = {
    'healthcareprofessional': 'healthcare professional',
    'practicestaff': 'practice staff',
    'missedappointments': 'missed appointments',
    'didnotattend': 'did not attend',
    'resourceutilization': 'resource utilization',
    'staffadequacy': 'staff adequacy',
    'budgetallocation': 'budget allocation',
    'resourceallocation': 'resource allocation',
    'utilisationtrends': 'utilisation trends',
    'noshowpatient': 'no show patient',
    'waitingtime': 'waiting time',
    'appointmentreminder': 'appointment reminder',
    'patientengagement': 'patient engagement',
    'socialimpact': 'social impact',
    'financialimpact': 'financial impact',
    'vulnerablecommunities': 'vulnerable communities',
    'punishingpatients': 'punishing patients',
    'investigatereasons': 'investigate reasons',
    'datainformeddecision': 'data informed decision',
    'expandcapacity': 'expand capacity',
    'existinginfrastructure': 'existing infrastructure',
    'betterutilization': 'better utilization',
    'reducemissedappointments': 'reduce missed appointments',
    'appointmentbooking': 'appointment booking',
    'appointmentmanagement': 'appointment management',
    'patientcommunication': 'patient communication',
    'healthcareaccess': 'healthcare access',
    'serviceutilization': 'service utilization',
    'reducenoshow': 'reduce no show',
    'increaseefficiency': 'increase efficiency',
    'resourceoptimization': 'resource optimization',
    'staffallocation': 'staff allocation',
    'healthservices': 'health services',
    'patientcare': 'patient care',
    'medicalappointments': 'medical appointments',
    'appointmentscheduling': 'appointment scheduling',
    'communityhealth': 'community health',
    'publichealth': 'public health',
    'healthcaresystem': 'healthcare system',
    'medicalpractice': 'medical practice',
    'primarycare': 'primary care',
    'extendedaccess': 'extended access',
    'virtualappointment': 'virtual appointment',
    'homevisit': 'home visit',
    'facetoface': 'face to face',
    'telephoneconsultation': 'telephone consultation',
    'videoconsultation': 'video consultation',
    'onlineconsultation': 'online consultation',
    'clinicaltriage': 'clinical triage',
    'plannedclinic': 'planned clinic',
    'medicationreview': 'medication review',
    'careencounter': 'care encounter',
    'unplannedactivity': 'unplanned activity',
    'carehomevisit': 'care home visit',
    'socialprescribing': 'social prescribing',
    'careplanning': 'care planning',
    'chargeablework': 'chargeable work',
    'groupconsultation': 'group consultation',
    'groupeducation': 'group education',
}

# Step 2: Expand the keyword list to include both concatenated and spaced versions
expanded_keywords = set(keywords)  # Start with the existing keywords

for concatenated, spaced in multi_word_keywords.items():
    expanded_keywords.add(spaced)  # Add the spaced version

# Convert the set back to a list
expanded_keywords = list(expanded_keywords)
```

```
# Step 2: Expand the keyword list to include both concatenated and spaced versions
expanded_keywords = set(keywords)  # Start with the existing keywords

for concatenated, spaced in multi_word_keywords.items():
    expanded_keywords.add(spaced)  # Add the spaced version

# Convert the set back to a list
expanded_keywords = list(expanded_keywords)

# Step 3: Lowercase the tweet_full_text column
tweets_exploded['tweet_full_text_lower'] = tweets_exploded['tweet_full_text'].str.lower()

# Step 4: Escape keywords to handle any special regex characters
keywords_escaped = [re.escape(word) for word in expanded_keywords]

# Step 5: Create regex patterns for each keyword
patterns = []
for keyword in expanded_keywords:
    # For multi-word phrases, wrap the phrase with word boundaries
    if ' ' in keyword:
        pattern = r'\b' + keyword + r'\b'
    else:
        # For single words, also use word boundaries
        pattern = r'\b' + keyword + r'\b'
    patterns.append(pattern)

# Combine all patterns into one regex pattern
keywords_pattern = '(' + '|'.join(patterns) + ')'

# Compile the pattern with case-insensitive matching
pattern_compiled = re.compile(keywords_pattern, re.IGNORECASE)
```

```
# Step 6: Define a function to find keywords in a text
def find_keywords(text):
    found_keywords = pattern_compiled.findall(text)
    # Normalize and remove duplicates
    found_keywords = [kw.lower() for kw in found_keywords]
    found_keywords = list(set(found_keywords))
    return found_keywords if found_keywords else None

# Step 7: Apply the function to create a new column with found keywords
tweets_exploded['found_keywords'] = tweets_exploded['tweet_full_text_lower'].apply(find_keywords)

# Step 8: Filter tweets that have at least one keyword found
tweets_filtered = tweets_exploded[tweets_exploded['found_keywords'].notnull()]

# Display the filtered DataFrame with the found keywords
tweets_filtered[['tweet_id','tweet_full_text','tweet_full_text_lower','found_keywords']]
```

Appendix 3

3.1

```
Pseudo Code:

1. Import necessary libraries:
   - Import `matplotlib.pyplot` as `plt` for plotting.
   - Import `seaborn` as `sns` for color palettes.

2. Process Data:
   - Group the data `ar` by "appointment_status" and sum the "count_of_appointments
   - Reset the index of the grouped data and store it in `ar_app`.

3. Define Color Palette:
   - Set a custom color palette in `palette_color` using specific color hex values.

4. Create Pie Chart:
   - Plot a pie chart using the `count_of_appointments` column from `ar_app`.
   - Use the "appointment_status" column as labels.
   - Apply the defined `palette_color` to the chart.
   - Set `autopct` to format percentages with no decimal places.
```

3.2

Pseudo Code:

1. Import necessary libraries:
   - Import `seaborn` as `sns` for data visualization aesthetics.
   - Import `matplotlib.pyplot` as `plt` for plotting.
   - Import `pandas` as `pd` for data manipulation.

2. Filter Data:
   - Select records in `ar` where "appointment_status" is either "Attended" or "DNA" and "appointment_mode" is not "Unknown".
   - Store the filtered data in `filtered_data`.

3. Group Data:
   - Group `filtered_data` by "appointment_mode" and "appointment_status" to get the count of appointments.
   - Sum the "count_of_appointments" for each group and store in `grouped_data`.

4. Calculate Totals for Each Mode:
   - Compute total appointments per "appointment_mode" to enable percentage calculation.
   - Store these totals in `total_per_mode`.

5. Calculate Percentage:
   - Calculate the percentage of each "appointment_status" within each "appointment_mode" and store in a new column `percentage` in `grouped_data`.

6. Pivot Data for Stacked Plot:
   - Pivot `grouped_data` to make "appointment_mode" the index, with "appointment_status" as columns.
   - Separate values into "count_of_appointments" and "percentage", filling missing values with zeroes.
   - Store the pivoted data in `pivot_data`.

7. Define Colors for Stacked Plot:
   - Set colors for each appointment status (green for "Attended" and red for "DNA").

8. Create Stacked Bar Chart:
   - Set up the plot dimensions with `fig` and `ax`.
   - Define `x` as the range of indices for the bars.

   - Plot "Attended" counts for each mode.
   - Stack "DNA" counts on top of "Attended" counts using `bottom` parameter.

9. Customize Plot:
   - Set the title, x-label, y-label, and x-tick labels for "appointment_mode" categories.

- Add a legend.

10. Annotate Percentages:
    - For each "appointment_mode," add percentage labels on each section of the stacked bars:
      - Place percentage annotations within "Attended" and above "DNA" segments where counts exist.

11. Adjust Layout:
    - Use `plt.tight_layout()` for better spacing.

12. Display Chart:
    - Use `plt.show()` to render the stacked bar chart.

3.3

Pseudo Code:

1. **Import Libraries:**
   - Import `seaborn` as `sns` for data visualization aesthetics.
   - Import `matplotlib.pyplot` as `plt` for plotting.
   - Import `matplotlib.dates` as `mdates` for date formatting on plots.
   - Import `pandas` as `pd` for data manipulation.

2. **Prepare Data:**
   - Convert the 'appointment_month_fr' column in `ar` DataFrame to datetime format.

3. **Filter Data:**
   - Select records where 'appointment_status' is either 'Attended' or 'DNA'.
   - Exclude records where 'appointment_mode' is 'Unknown'.
   - Store the filtered data in `ar_grp`.

4. **Group and Aggregate Data:**
   - Group `ar_grp` by 'appointment_month_fr' and 'appointment_status'.
   - Sum the 'count_of_appointments' for each group and reset the index.
   - Store the result in `ar_grp_count`.

5. **Calculate Totals and Percentages:**
   - For each 'appointment_month_fr', calculate the total number of appointments.
   - Compute the percentage of 'Attended' and 'DNA' appointments per month.
   - Add a new column 'percentage' to `ar_grp_count` with these values.

6. **Pivot Data for Plotting:**
   - Create a pivot table `pivot_percent` with 'appointment_month_fr' as the index and 'appointment_status' as columns, containing percentage values.

- Create another pivot table `pivot_counts` with the same index and columns, containing count values.
- Fill any missing values with zero.

7. **Define Plot Aesthetics:**
- Set colors for 'Attended' and 'DNA' statuses (e.g., green and red).

8. **Create Combined Plot with Dual Axes:**
- Initialize a figure and primary axis (`ax1`) with specified size.
- Create a secondary axis (`ax2`) that shares the same x-axis as `ax1`.

9. **Plot Stacked Area Chart on Secondary Axis:**
- Use `stackplot` to plot 'Attended' and 'DNA' percentages over time on `ax2`.
- Set labels, colors, and transparency (alpha) for the stacked areas.
- Label the y-axis for percentages and set its limits from 0 to 100%.
- Add a legend to `ax2`.

10. **Plot Line Charts on Primary Axis:**
- For each 'appointment_status' in `pivot_counts`, plot a line representing the count of appointments over time on `ax1`.
- Add markers to the lines for better visibility.
- Set the plot title and axis labels.
- Add a legend to `ax1`.

11. **Format X-Axis:**
- Set major ticks to display every month using `MonthLocator`.
- Format the x-axis labels to show year and month (e.g., '2023-01').
- Rotate x-axis labels by 45 degrees for readability.

12. **Finalize and Display Plot:**
- Adjust the layout for better spacing using `plt.tight_layout()`.
- Render the combined plot using `plt.show()`.

Appendix 4

4.1

Pseudo Code:

1. **Import Libraries:**
   - Import the `pandas` library as `pd` for data manipulation.
   - Import the `datetime` module for handling date and time operations.

2. **Split Hashtags into a List:**
   - Take the `tweet_entities_hashtags` column from the `tweets_fil` DataFrame.
   - Split each entry by commas to create a list of hashtags.

3. **Expand Hashtags to Individual Rows:**
   - Use the `explode` function on the `hashtags_list` column to create a separate row for each hashtag.
   - Store the expanded DataFrame in `tweets_exploded`.

4. **Standardize Hashtags:**
   - Convert all hashtags in the `hashtags_list` column to lowercase to ensure consistency.
   - Remove any leading or trailing whitespace from the hashtags.
   - Store the standardized hashtags in a new column called `hashtags_list_lower`.

5. **Define Function to Extract Date from Tweet ID:**
   - Create a function `get_date_from_tweet_id` that takes a `tweet_id` as input.
     a. Convert `tweet_id` to an integer.
     b. Define the Twitter epoch in milliseconds (`1288834974657`).
     c. Shift the `tweet_id` right by 22 bits to remove the lower bits.
     d. Add the shifted `tweet_id` to the Twitter epoch to get the timestamp in milliseconds.
     e. Convert the timestamp from milliseconds to seconds.
     f. Create a `datetime` object from the timestamp.
     g. Extract and return the date part of the `datetime` object.

6. **Apply Date Extraction Function to Get Posting Dates:**
   - Apply the `get_date_from_tweet_id` function to the `tweet_id` column of the `tweets_exploded` DataFrame.
   - Store the resulting dates in a new column called `Date`.

4.2

Pseudo Code:

1. **Import Necessary Libraries:**
- Import the `nltk` library for natural language processing.
- Import the `re` module for regular expression operations.
- Import the `datetime` module for handling date and time.
- Import `pandas` as `pd` for data manipulation.

2. **Download VADER Lexicon:**
- Use `nltk.download('vader_lexicon')` to download the VADER lexicon required for sentiment analysis.

3. **Initialize Sentiment Analyzer:**
- Import `SentimentIntensityAnalyzer` from `nltk.sentiment.vader`.
- Create an instance of `SentimentIntensityAnalyzer` and store it in `sia`.

4. **Define Sentiment Analysis Function:**
- Create a function `get_sentiment(text)` that:
  - Uses VADER's `polarity_scores` method to analyze the input `text`.
  - Returns a dictionary containing sentiment scores: 'neg', 'neu', 'pos', and 'compound'.

5. **Define Text Preprocessing Function:**
   - Create a function `preprocess_text(text)` that:
     - Removes URLs using a regular expression pattern matching 'http' or 'www'.
     - Removes mentions (e.g., @username) and hashtags (e.g., #hashtag) using a regular expression.
     - Optionally removes special characters and numbers, retaining only letters and whitespace.
     - Strips leading and trailing whitespace from the cleaned text.
     - Returns the cleaned text.

6. **Apply Text Preprocessing:**
   - Apply the `preprocess_text` function to the `tweet_full_text` column of the `tweets_exploded` DataFrame.
   - Store the cleaned text in a new column called `clean_text`.

7. **Perform Sentiment Analysis:**
   - Apply the `get_sentiment` function to the `clean_text` column of the `tweets_exploded` DataFrame.
   - Store the resulting sentiment scores in a new column called `sentiment_scores`.

8. **Extract Compound Sentiment Score:**
   - From the `sentiment_scores` column, extract the 'compound' score using a lambda function.

9. **Define Sentiment Classification Function:**
   - Create a function `classify_sentiment(compound_score)` that:
     - Returns 'positive' if `compound_score` is greater than or equal to 0.05.
     - Returns 'negative' if `compound_score` is less than or equal to -0.05.
     - Returns 'neutral' otherwise.

10. **Classify Sentiment:**
    - Apply the `classify_sentiment` function to the `compound_score` column of the `tweets_exploded` DataFrame.
    - Store the classification results in a new column called `sentiment`.